THE ECONOMIC JOURNAL

The Economic Journal, 1–32 https://doi.org/10.1093/ej/ueaf050 © The Author(s) 2025. Published by Oxford University Press on behalf of Royal Economic Society. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com.
Advance Access Publication Date: 20 June 2025

CLIMATE SHOCKS, CYCLONES AND ECONOMIC GROWTH: BRIDGING THE MICRO-MACRO GAP*

Laura A. Bakkensen and Lint Barrage

This paper proposes a joint empirical-structural approach to quantify the impacts of cyclone risks on economic growth and welfare. First, we review prior reduced-form approaches in a harmonised global dataset and through a theory lens. Second, we estimate cyclone impacts on structural determinants of growth (total factor productivity, capital losses and fatalities) to quantify a stochastic endogenous growth cyclone-economy model for forty vulnerable countries. Third, we study climate change impacts on cyclone risk. The results suggest negative but mostly modest impacts of risk changes on growth, which may nonetheless add up over time to substantial welfare losses, especially in the Caribbean and the United States.

JEL codes: O44, O47, Q54

What are the macroeconomic consequences of climate change? A rapidly growing body of empirical work has associated weather events such as temperature anomalies and tropical storms with large impacts on aggregate output. While many of these studies seek to inform estimates of the social costs of climate change, their findings have often been slow to be incorporated into macroeconomic climate-economy models. On the one hand, there are still fundamental questions about the methodologies and interpretation of this empirical work (see, e.g., Newell *et al.*, 2021; Carleton and Greenstone, 2022). Indeed, for tropical storms, the empirical literature has found a wide range of results, including positive (e.g., Skidmore and Toya, 2002), mixed (e.g., Noy, 2009), small negative (e.g., Strobl, 2011) and very large negative (e.g., Hsiang and Jina, 2014) effects on growth. On the other hand, it is often unclear how to map reduced-form evidence into structural models. Given that climate-economy models are used to quantify the social cost of carbon emissions (Greenstone *et al.*, 2013), and given that empirically estimated output impacts could add significantly to these costs (e.g., Moore and Diaz, 2015), this 'micro-macro' gap represents both an important academic and policy concern.

* Corresponding author: Laura A. Bakkensen, Department of Economics, University of Oregon, 175 Prince Lucien Campbell Hall, Eugene, OR 97403-1205, USA. Email: lbak@uoregon.edu

This paper was received on 11 December 2019 and accepted on 18 June 2025. The Editor was Estelle Cantillon.

The data and codes for this paper are available on the Journal repository. They were checked for their ability to reproduce the results presented in the paper. The authors were granted an exemption to publish parts of their data because access to these data is restricted. However, the authors provided the Journal with temporary access to the data, which enabled the Journal to run their codes. The codes for the parts subject to exemption are also available on the Journal repository. The restricted access data and these codes were also checked for their ability to reproduce the results presented in the paper. The replication package for this paper is available at the following address: https://doi.org/10.5281/zenodo.15627143.

We thank Estelle Cantillon and two anonymous referees for their insightful comments and feedback. We also thank Kerry Emanuel and Wind Risk Tech, LLC for the simulated cyclone track data utilised in this analysis. We thank Tatyana Deryugina, John Acknowledgements and Rodolfo Manuelli for insightful discussions of this paper, and we thank Steve Cicala, Stephane Hallegatte, Sol Hsiang, William Nordhaus, Aleh Tsyvinski, Mattew Turner, Kieran Walsh, David Weil and seminar participants at UC Berkeley, Yale FES, NBER EEE Summer Institute, IMF Workshop on Macroeconomic Policy and Inequality, Colby College, University of Rhode Island, SURED 2018, SED 2018, the Federal Reserve Board and the University of Chicago (Harris School) for their helpful feedback and comments.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

This paper proposes a novel joint empirical-structural approach to overcome this gap in the context of tropical cyclones (i.e., hurricanes, typhoons). Cyclones are the leading cause of natural disaster damage worldwide.[1] A large empirical literature has investigated the impacts of tropical cyclones on economic growth (e.g., Skidmore and Toya, 2002; Noy, 2009; Raddatz, 2009; Strobl, 2011; Loayza *et al.*, 2012; Fomby *et al.*, 2013; Hsiang and Jina, 2014; McDermott *et al.*, 2014; Elliott *et al.*, 2015; see also the review by Kousky, 2014). Importantly, cyclone impacts are projected to increase in the coming decades due to global changes (e.g,. Nordhaus, 2010b; Mendelsohn *et al.*, 2012; Ranson *et al.*, 2014). Our contribution to this literature is two-fold.

First, we review competing prior empirical approaches to quantifying cyclone impacts on economic growth. We assemble a modern dataset encompassing meteorological measures of the full history of cyclones around the globe from 1970–2015 in order to revisit prior approaches in a harmonised sample. On the one hand, we find that key divergent prior findings can be reconciled based on methodological differences in estimation. On the other hand, through the lens of macroeconomic theory, we also show that prior reduced-form approaches cannot capture the full impacts of future cyclone risk changes on growth and welfare. For example, panel fixed-effect models may isolate the direct output losses from cyclone strikes, but fail to capture long-run and indirect impacts of cyclone risk on margins such as savings and investment behaviour.

Second, we propose a novel approach that combines empirical estimation with the structure of a model to account for both direct and indirect impacts and to quantify the welfare costs of cyclone risk. On the empirical side, we first quantify cyclone impacts on the structural determinants of growth, rather than on growth itself. Specifically, we estimate cyclone strike impacts on total factor productivity (TFP), capital destruction and fatalities within our global panel. On the modelling side, we present a stochastic endogenous growth cyclone-climate-economy model. Our framework builds closely on that of Krebs (2003a,b; 2006; Krebs et al. 2015), who studied the implications of business cycles and idiosyncratic human capital risks for growth and welfare. One key feature of the model is that it makes an explicit distinction between climate and weather: households face repeated risks to their physical and human capital from cyclone strikes (weather), whose probability distribution is determined by the climate. This setup enables us to directly calibrate cyclone impacts to the plausibly causally identified estimates from panel (weather) regressions. In addition, the model structure enables us to account for household responses to changing cyclone risks (climate) through changes in savings and asset allocations. Our approach thus meets the three goals for modern climate impact quantification as proposed by Greenstone (2016) and Auffhammer (2018): (i) plausibly causally identified empirical impact estimates, (ii) accounting for adaptation through endogenous adjustments in savings and investments, and (iii) computing welfare costs of changes in climatic risks.

We quantify our cyclone-climate-economy model separately for each of the forty cyclone-vulnerable nations. The calibration includes country-specific probability distributions of current and future cyclone intensity that we estimate based on 68,000 synthetic storm track simulations from Emanuel *et al.* (2008) and historical best track cyclone data. We use the model to quantify the growth and welfare impacts of both present-day and future cyclone risk distributions.

Our main finding is that cyclone risk changes from global warming are projected to impose significant welfare costs on many countries. Even under a moderate emissions scenario, welfare losses are estimated to reach up to a 5% balanced growth path consumption equivalent loss in

[1] Comparing overall natural event losses worldwide (1998–2008) from cyclones to earthquakes/tsunamis, convective storms, winter storms, floods and heatwaves/fires in MunichRe's NatCatService database.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

the most vulnerable countries (e.g., St. Vincent and the Grenadines). Interestingly, though the most negatively impacted countries are generally poor and/or small island states (e.g., Belize, Haiti, Comoros, etc.), the United States is also among the top ten impacted countries. This result is informed by prior empirical work identifying the United States as an outlier in cyclone vulnerability conditional on its income levels and exposure (e.g., Bakkensen and Mendelsohn, 2016), and highlights the importance of work investigating the determinants of adaptation to storms in the United States (e.g., Fried, 2022).

We also find negative impacts of cyclone risk on average economic growth. This finding is distinct from the negative impacts of cyclone strikes on short-run growth that has been documented empirically, and that our model replicates. Present-day cyclone risks are estimated to decrease average annual growth rates by between 0.01 and 0.1 percentage points in the ten most vulnerable countries in our sample. One reason why these figures may appear smaller than some prior empirical estimates is that our model accounts for both direct and indirect effects of storms. Indeed, we find that behavioural adjustments to changes in storm risk can mitigate their negative growth impacts by up to 30%–40%.

Our analysis complements several recent literature advancements on the quantitative modelling of natural disasters and the macroeconomy. Perhaps most closely related in methodology, Fried (2022) presented a dynamic general equilibrium model of the US economy where heterogeneous households face risks of capital destruction from storms. She presented an original calibration using US Federal Emergency Management Agency (FEMA) disaster assistance across regions, and utilised the model to quantify adaptation capital, FEMA policy effects and the role of adaptation in mitigating welfare costs from future storm intensity increases. Hallegatte *et al.* (2007) developed a 'non-equilibrium dynamic model' (NEDyM) of disasters and applied it to extreme weather events (albeit not cyclones)[2] in Europe. NEDyM builds on a Solow growth model with limits on reconstruction investment and non-clearing short-run labour and goods markets. As both our model structure and research goals are fundamentally different, we abstract from some of the important nuances featured in these studies, and focus instead on detailed linkages to empirical reduced-form approaches and long-run growth impacts under aggregate risk.[3,4] Conceptually, our analysis is also informed by a vast literature in macroeconomics on the effects of uninsurable risks on economic growth and welfare (e.g., Bewley, 1977; Lucas, 1987; Aiyagari, 1994; Krebs, 2003b). In recent years, several theoretical studies have considered natural disaster risks and growth in particular (e.g., Ikefuji and Horii, 2012; Akao and Sakamoto, 2013; Müller-Fürstenberger and Schumacher, 2015; Bretschger and Vinogradova, 2018).[5]

More broadly, our analysis also relates to a parallel literature on temperature shocks and economic growth. Several influential empirical studies have argued that temperature shocks affect

[2] Hallegatte (2009) combined empirical direct cyclone impact estimates for the United States with estimates of the relationship between direct and indirect losses (based on a case study applying an input-output model to Hurricane Katrina's sectoral impacts) to project total economic losses from hurricanes in the United States both with and without climate change-induced hurricane intensity increases.

[3] For example, besides structural differences, Fried (2022) featured no aggregate risk from storms in the United States. In contrast, we study aggregate growth impacts of cyclone risk across forty vulnerable countries. Similarly, Hallegatte *et al.* (2007) focused on short-run transitional additions to overall disaster costs. In contrast, our focus is on long-run growth effects through changes in asset allocations and savings rates, which are exogenous in NEDyM.

[4] Another related study by Narita *et al.* (2009) used the FUND integrated assessment model to project direct cyclone damage impacts on the social cost of carbon, but not on the macroeconomy.

[5] Some studies have considered the macroeconomic impacts of rare disaster risk (e.g., Barro, 2006; Pindyck and Wang, 2013). Cyclones are typically not examples of such rare disasters, however, as they are both common in many countries and physically limited (Holand and Emannuel, 2011). Pindyck and Wang (2013) defined catastrophic shocks as reducing capital by 'more than 10 or 15%'. In our data, even the 95th percentile of capital destruction is only 2.8%.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

economic growth (e.g., Bansal and Ochoa, 2011; Dell *et al.*, 2012; Burke *et al.*, 2015; Colacito *et al.*, 2019). Though the robustness and interpretation of these results remains subject to an active debate (e.g., Kalkuhl and Wenz, 2020; Newell *et al.*, 2021), there have been efforts to incorporate these findings into climate-economic models. One pioneering analysis by Moore and Diaz (2015) seeks to incorporate the findings of Dell *et al.* (2012) into the seminal DICE model (Nordhaus, 2017). As output growth impact estimates do not provide a clear mapping into macroeconomic models, Moore and Diaz (2015) considered calibrating either TFP growth or capital depreciation to match reduced-form estimates. While both mechanisms lead to an increase in the social cost of carbon, the choice between them appears quantitatively significant.[6] A similar issue arises in the work of Fankhauser and Tol (2005) and Dietz and Stern (2015), who extended DICE to an endogenous long-run growth framework with capital- or investment-based knowledge spillovers. Absent empirical guidance, they also considered a capital depreciation specification and a TFP depreciation specification. Notably, the optimal carbon price in 2015 is 55% higher with the TFP specification, again highlighting the critical importance of understanding impact channels. In order to overcome these ambiguities, we propose a joint empirical-structural approach that directly estimates cyclone impacts on the determinants of growth.[7,8] While the mechanisms through which cyclones affect the economy are of course different than those associated with temperature fluctuations, our approach relates methodologically to this broader literature.

The remainder of this paper proceeds as follows. Section 1 describes our data and presents our review of reduced-form approaches to quantifying cyclone impacts on economic growth. Section 2 presents our stochastic endogenous growth cyclone-climate-economy model. Section 3 details our quantification of the model, including the empirical estimation of cyclone strike impacts on TFP, capital destruction and fatalities and our estimation of cyclone risk probability distributions. Finally, Section 4 presents our modelling results, and Section 5 concludes.

1. Revisiting Empirical Approaches

1.1. *Data*

The first step in our analysis is to compile a harmonised global panel of cyclones and relevant economic indicators at the country-year level.

1.1.1. *Cyclone data*

Building on best practices in the literature (Hsiang and Jina, 2014), we gather historical global tropical cyclone tracks from the International Best Track Archive for Climate Stewardship (IBTrACS; Knapp *et al.*, 2010; Knapp, 2018). Considered the most comprehensive record of global historical tropical cyclone tracks by the World Meteorological Organization, IBTrACS contains

[6] Moore and Diaz (2015) ultimately focused on the TFP pathway. Supplementary results for the depreciation pathway suggest broadly similar patterns but a significantly higher social cost of carbon. Gauging visually from the relevant graphs, the Social Cost of Carbon appears to reach well over $1,500+ per ton by 2080 for depreciation damage, compared to around $1,000 per ton for the TFP pathway in the benchmark.

[7] Alternative approaches include, e.g., Bansal and Ochoa (2011), who presented a long-run risk model calibrated to their own estimates of temperature shock growth impacts. Consumption growth is a given process in this model; that is, it is not a production-based growth model, thus side-stepping questions of impact mechanisms.

[8] A growing number of studies present climate-economy models with growth impacts but without focus on empirical connections. Bretschger and Valente (2011) provided a theoretical foundation for climate change growth impacts through multiple channels (capital and TFP depreciation). Lemoine (2021) presented an endowment economy with temperature impacts on consumption growth to study the implications of uncertainty for the social cost of carbon. Of course, there is also a broader theoretical literature on the environment and endogenous growth (e.g,. Bovenberg and Smulders, 1995).

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

best track records of cyclone position and intensity characteristics collected from meteorological agencies across the world. We focus on 1970–2015, the post-satellite era for which cyclones have been most reliably tracked. For all 3,346 cyclone landfalls during this period, we calculate cyclone intensity metrics including annual maximum wind speed at landfall (in knots) and annual energy (the sum of cubes of wind speeds recorded within a country), a metric based on the power dissipation index developed by Emanuel (2008).[9] We process the tracks in ArcGIS and aggregate data up to the country-year level.[10] Importantly, we divide all meteorological cyclone intensity measures by country land areas in order to account for potential heterogeneity in macroeconomic relevance of cyclones for countries of different sizes. Next, in order to estimate future changes in cyclone risks, we incorporate 68,000 simulated future tropical cyclone tracks based on advancements in climatological research by Kerry Emanuel and co-authors (Emanuel, 2008; Emanuel *et al.*, 2008). These synthetic tracks and their usage are described in detail in Section 3.3 below.

1.1.2. *Macroeconomic indicators*

We collect annual national-level macroeconomic indicators including real GDP (2011 $US), physical and human capital stocks, and population from the Penn World Tables 9.0 ('PWT'; Feenstra *et al.*, 2015). Though not directly used in our analysis, we further obtain World Bank data (from World Development Indicators; WBWDI, 2019) on several other macroeconomic indicators (gross capital formation and imports, foreign direct investment and government surplus, all as percentage of GDP) included in analyses such as Noy (2009) in order to construct a comparable sample based on data availability (described in Section 1.2 below).

1.1.3. *Geography*

Country areas and absolute latitudes are collected from the Harvard WorldMap. We also collect information on countries' populations living below five metres of elevation from the Low Elevation Coastal Zone Urban-Rural Population and Land Area Estimates from the Center for International Earth Science Information Network (CIESIN, 2013) at Columbia University and as published through the Socioeconomic Data and Applications Center.

1.1.4. *Institutions*

In line with the broader disaster literature (e.g., Noy, 2009; McDermott *et al.*, 2014), we consider the World Bank's measure of domestic credit provided by the financial sector (as a percentage of GDP) as a proxy for financial market development. We also collect World Bank data on countries' 'Statistical Capacity Ratings'.

1.1.5. *Cyclone damage*

Finally, we obtain cyclone damage estimates from two sources. Our benchmark measure of property damage and fatalities is gathered from EMDAT, the International Disaster Database (Guha-Sapir *et al.*, 2016). EMDAT is the most comprehensive publicly available database on disaster losses and arguably the most widely used in the literature (e.g., Skidmore and Toya,

[9] Given that some cyclone wind speeds are listed as zero while a cyclone necessarily has non-zero wind speeds, we interpolate missing wind speeds from minimum pressure readings following Atkinson and Holliday (1977). For a minority of observations missing both wind and pressure, we assume a wind speed of 35 knots for categorised cyclones and 25 knots for tropical depressions. Lastly, we convert one-minute sustained wind speeds to ten-minute sustained wind speeds for unit consistency.

[10] We process the data without a dedicated wind-field model. For recent advancements on such modelling, see, e.g., Strobl (2011), Hsiang and Narita (2012), Hsiang and Jina (2014).

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

2002; Raddatz, 2007; Noy, 2009; Hsiang and Narita, 2012). At the same time, EMDAT data are subject to certain data quality caveats (e.g., Hsiang and Narita, 2012). While comparative analyses with proprietary damage data from global re-insurance companies fail to indicate that these would necessarily dominate EMDAT data coverage (Guha-Sapir and Below, 2002), for robustness, we also consider damage estimates from MunichRe. We specifically use country-year aggregates of total direct losses from cyclones as computed by Neumayer *et al.* (2014) from the MunichRe database.

1.2. *Estimation*

The most common empirical approach to studying natural disaster impacts on growth is to use panel variation. Conceptually, this approach captures the impact of cyclone *strikes* on realised growth. This literature has documented a range of results, with most finding negative effects of varying magnitude and duration (see, e.g., Kousky, 2014), but some finding no impacts in all but the most extreme disasters (Cavallo *et al.*, 2013).[11] As a first step, we thus consider a panel specification similar to Hsiang and Jina (2014) in our harmonised global sample:

$$g_{j,t} = \gamma_j + \delta_t + (\theta_j \cdot t) + \sum_{l=0}^{L} \beta_{1+l}\varepsilon_{j,t-l} + \beta_{Int}(q_{j,t} \cdot \varepsilon_{j,t}) + \epsilon_{j,t}. \tag{1}$$

Here, $g_{j,t}$ is a country's annual real GDP per capita growth rate, the γ_j are country fixed effects, the $(\theta_j \cdot t)$ are country-specific linear time trends and the $\varepsilon_{j,t-l}$ are cyclone realisation measures up to lag L. Our preferred cyclone intensity measure $\varepsilon_{j,t}$ is maximum sustained cyclone wind speed per square kilometre of land area. However, parts of our analysis also consider the number of cyclone landfalls per land area or cyclone energy (the sum of cubes of maximum wind speeds in a given country-year pair). Online Appendix Table A0 presents summary statistics for each of these measures. Our output growth panel regressions focus on contemporaneous impacts ($L = 0$), but we consider richer lag structures in our main empirical impact channel estimation in Section 3 below. The empirical literature has frequently found that disaster impacts vary with country characteristics, particularly the level of development and the quality of (financial) institutions (e.g., Kahn, 2005; Noy, 2009; Raddatz, 2009; Loayza *et al.*, 2012; Fomby *et al.*, 2013; McDermott *et al.*, 2014). Specification (1) consequently allows for the impact of cyclones to vary with covariates $q_{j,t}$, specifically domestic credit or lagged GDP per capita.[12] SEs $\epsilon_{j,t}$ are heteroscedasticity robust and clustered at the country level.

In reviewing the literature, we document another potential source of variation in results: sample composition based on macroeconomic control variable availability. That is, prior studies differ in the control variables they consider. One classic example, Noy (2009), includes a rich set of controls such as government budget surplus and foreign exchange reserves, permitting a final sample of 109 countries. Numerous other recent studies end up with similar or smaller samples,[13] whereas,

[11] Prior research has already identified study design features that can contribute to differences in results. For example, Loayza *et al.* (2012) documented heterogeneous impacts across disaster types (e.g., earthquakes versus storms). Different proxies for disaster intensity (e.g., property damage versus fatalities) have also been shown to yield different results (Noy, 2009). Given our focus on one disaster type, cyclones, and one intensity measure type, meteorological, these factors cannot account for differences in results in our setting.

[12] We lag GDP to avoid endogeneity to the year t disaster realisation, but consider contemporaneous credit as it reduces impacts precisely through its response to disasters.

[13] For example, Loayza *et al.* (2012) had a sample of ninety-four countries and Fomby *et al.* (2013) studied eighty-four countries.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Table 1. *Panel Analysis: Cyclone Strikes and Growth.*

Dependent variable:	Real GDP/Capita Growth$_{j,t}$					
	Unfiltered			Has Controls		
	(1)	(2)	(3)	(4)	(5)	(6)
#Landfalls/sqkm$_{j,t}$	0.637	−1.966	−19.392	−67.303**	−474.390**	−1,240.100***
	(1.282)	(20.517)	(33.566)	(29.006)	(191.384)	(431.845)
Credit$_{j,t}$ · (#Landfalls/sqkm$_{j,t}$)		0.010			5.394**	
		(0.273)			(2.672)	
ln (GDP p.c.)$_{j,t-1}$ · (#Landfalls/sqkm$_{j,t}$)			1.961			118.624***
			(3.308)			(42.363)
Domestic Credit$_{j,t}$		−0.000			−0.000**	
		(0.000)			(0.000)	
ln (GDP p.c.)$_{j,t-1}$			−0.103***			−0.220***
			(0.013)			(0.033)
Adj. R^2	0.110	0.0985	0.165	0.177	0.201	0.277
Max. Wind/sqkm$_{j,t}$	0.004	−0.890	−2.699	−2.280***	−3.738**	−4.744
	(0.050)	(1.193)	(2.059)	(0.299)	(1.833)	(10.105)
Credit$_{j,t}$ · (Max. Wind/sqkm$_{j,t}$)		0.010			0.020	
		(0.015)			(0.023)	
ln (GDP p.c.)$_{j,t-1}$ · (Max. Wind/sqkm$_{j,t}$)			0.274			0.262
			(0.207)			(1.008)
Domestic Credit$_{j,t}$		−0.000			−0.000**	
		(0.000)			(0.000)	
ln (GDP p.c.)$_{j,t-1}$			−0.103***			−0.219***
			(0.013)			(0.033)
Adj. R^2	0.110	0.0993	0.166	0.178	0.200	0.278
Observations	7,573	5,690	7,573	1,978	1,978	1,978
#Countries	182	171	182	116	116	116
Country FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Country trends	Yes	Yes	Yes	Yes	Yes	Yes
SE cluster	Country	Country	Country	Country	Country	Country

Notes: This table presents the regression of countries' real GDP per capita growth rate in year t on the number of cyclone landfalls per square kilometre (top panel) or the maximum wind speed per square kilometre in year t plus controls for the lagged natural log of real GDP per capita in level and interacted with storms (columns (3) and (6)) or domestic credit provided by the financial sector (%GDP) in level and interacted with storms (columns (2) and (5)). All regressions include country fixed effects, year fixed effects, country-specific linear time trends and a constant. SEs are heteroscedasticity robust and clustered at the country level. *** $p < .01$, ** $p < .05$.

e.g., Cavallo *et al.* (2013) used a synthetic control approach and constructed certain variables permitting a sample of 196 countries. In order to gauge whether such sample differences may be contributing to differences in studies' results, we estimate (1) for two samples: (i) 'Unfiltered' includes all available countries (182 countries), whereas (ii) 'Has Controls' includes only country-year pairs for which control variables used by Noy (2009) and others are available.[14] We do *not* actually include those controls in the regressions so as to isolate sample effects.

Table 1 presents the results for landfall counts and maximum wind speed per square kilometre as intensity metrics. Results for energy are presented in Online Appendix Table A1. First, the

[14] We specifically define a sample of country-year pairs that have data on gross capital formation, domestic credit, imports as percentage of GDP, foreign direct investment, government surplus and countries that have at any point had institutional quality ratings from the *International Country Risk Guide* (ICRG). This sample does *not* match Noy's exactly due to changes in source databases over time. We also do not purchase the *ICRG* data and utilise public metadata instead (https://epub.prsgroup.com/available-countries). If countries have entered this database since the time of Noy''s analysis, they may also change the relative samples. Importantly, our goal is not to replicate Noy's sample per se, but to demonstrate how a representative example of standard controls can affect the sample precision and results.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

results generally confirm that cyclone *strikes* have a negative effect on contemporaneous economic growth. Second, the statistical precision of these results differs notably across the unfiltered and data-restricted country samples. In the unfiltered sample, contemporaneous cyclone impacts are precisely estimated only for energy, whereas they are generally significantly different from zero in the data-restricted sample, broadly in line with some of the underlying heterogeneity in the literature.[15] As shown in Online Appendix Table A2, countries in the restricted sample feature significantly higher average statistical capacity, lower volatility in GDP growth and larger average populations than the unfiltered sample, likely contributing to the difference in precision of the estimated results. Finally, in line with prior studies, we see that these negative growth impacts are generally lower in countries with better financial institutions (proxied by domestic credit; columns (2) and (5)) and higher levels of development (columns (3) and (6)).

An alternative approach to analysing the association between natural disasters and economic growth has been cross-sectional analysis. Though less common, this approach is used in one of the most widely cited papers in the literature by Skidmore and Toya (2002).[16] Their study regresses countries' average 1960–90 growth rates on disaster metrics such as the average number of climatic events per year. In stark contrast to the panel literature, Skidmore and Toya (2002) found a *positive* correlation between disasters and growth. We show in the Online Appendix that this surprising result survives in our harmonised modern dataset.[17] That is, in the same global analysis that yields a negative panel impact estimate of cyclone strikes on growth, we find a positive correlation between average cyclone risk and growth. We next describe potential reasons for this divergence and the shortcomings of both panel and cross-sectional methods for quantifying the full impacts of cyclone risk changes.

1.3. *Discussion*

From a macro-theoretical perspective, it is not surprising that panel and cross-sectional analyses produce such different results. Panel specifications such as (1) capture the impact of cyclone *strikes* on realised growth, whereas cross-sectional regressions intend to capture the effect of cyclone *risk* on average long-run growth, although omitted variable bias is of course a major concern. Conceptually, cyclone strikes should have a direct negative impact on GDP growth if they destroy productive assets or disrupt production processes. As households and firms respond to the risk of such strikes, however, several indirect mechanisms may lead to a different relationship between cyclone risk and long-run growth. For example, an increase in economic risk may induce households to save more, that is, to undertake precautionary savings (e.g., Bewley, 1977). Ceteris paribus, an increase in savings rates can increase average growth rates across a range of macroeconomic models.[18] As another example, to the extent that cyclone risk alters

[15] For example, studies such as Noy (2009) found significant negative effects of disasters on growth, whereas Cavallo *et al.* (2013) did not except in the largest disasters. Of course, there are many other methodological differences across these and other studies, and it is beyond the scope of our study to formally decompose differences in results across these factors.

[16] Another cross-sectional study by Hsiang and Jina (2015) documents a negative cross-sectional relationship between average cyclone-induced capital depreciation and growth.

[17] This supplementary analysis uses additional data to control for education (Barro and Lee, 2013), institutions (CPI, 2015), tropical area shares (Rubel and Kottek, 2010) and water proximity (Wessel and Smith, 1996).

[18] For example, an increase in savings rates could increase growth (i) in a Solow growth model during the transition to a long-run balanced growth path, (ii) in an endogenous growth model with aggregate capital externalities (see, e.g., Devereux and Smith, 1994), (iii) in an AK-type endogenous growth model (see, e.g.,Krebs, 2006) and (iv) in a Lucas (1988)-style model of human capital-driven growth (see, e.g., Ikefuji and Horii, 2012).

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

the relative attractiveness of different investment options (e.g., physical versus human capital), it may also alter long-run growth indirectly by changing the *composition* of an economy's assets. Importantly, these considerations suggest that neither panel nor cross-sectional growth regressions are sufficient to quantify the full impacts of future changes in cyclone risks. On the one hand, while panel approaches are clearly attractive in terms of econometric identification, they fail to account for the indirect effects of cyclone risk on long-run growth.[19] Cross-sectional approaches, on the other hand, are vulnerable to omitted variable bias (see, e.g., the discussion in Auffhammer, 2018). More fundamentally, growth impacts also need not be informative about welfare. It is long known that changes in economic risk can affect growth and welfare in opposite ways (see, e.g., Devereux and Smith, 1994). For example, if higher cyclone risk increases growth by increasing precautionary savings, this change is clearly welfare reducing. Consequently, even a perfectly identified cross-sectional growth regression would not be sufficient to quantify welfare impacts of cyclone risk changes. In sum, these results thus highlight the need for a structural approach to capture both the welfare and general equilibrium effects of climate-induced changes in future cyclone risks. The next section describes our framework.

Before proceeding, we note that panel output growth regressions do provide essential insights that can inform the design of environment-economy models. For example, limited financial markets are clearly an empirically relevant contributor to vulnerability, but not accounted for in many integrated assessment models. Another common empirical finding is that negative cyclone strike impacts on output *levels* appear *persistent* (e.g., Raddatz, 2007; Strobl, 2011; Hsiang and Jina, 2014; Elliott *et al.*, 2015). This stylised fact is at odds with a standard Ramsey model, which would imply a growth rebound after the initial negative impact, motivating our exploration of an alternative framework.

2. Stochastic Endogenous Growth Cyclone-Economy Model

This section presents our stochastic endogenous growth cyclone-economy model. The goal of the model is to both incorporate empirical estimates of the direct impacts of storms and to account for indirect impacts of cyclones on growth and welfare. Our framework builds closely on that of Krebs (2003a,b; 2006; Krebs et al. 2015).[20] We model each country as a closed economy where agents can invest in human and physical capital. Cyclone strikes can damage both types of capital as well as TFP. Intuitively, human capital damage may represent mortality, other physical or mental health impacts and disruptions to education. TFP impacts could likewise result from a variety of channels such as electricity outages (Allcott *et al.*, 2016), labour displacement (Tierney, 1997) or complementarities among heterogeneous types of capital (Hallegatte and Vogt-Schilb, 2019), as discussed further in Section 3 below. Households respond to the risk of cyclones through their savings and investment decisions. Importantly for our purposes, the model thus enables us to represent both direct and indirect cyclone impact channels and to reproduce some key results

[19] Formally, the fixed effects related to countries' average growth rates in (1), γ_j and θ_j, are endogenous to cyclone *risk*. This endogeneity becomes important if one wishes to use panel estimates to predict climate change impacts. That is, while some empirical studies have analysed climate change impacts by evaluating (1) at alternative potential future storm *realisations* (e.g., Hsiang and Jina, 2014), this approach does not account for climate change impacts on average growth rates due to changes in expected cyclone risk. A prior version of this paper explored a two-step estimator to evaluate cyclone impacts through both channels (Bakkensen and Barrage, 2016).

[20] Krebs developed a heterogeneous agent version of this class of model to study the implications of idiosyncratic human capital and business cycle risks for household savings, investment, growth and welfare. We consider a representative agent economy, but allow for (i) correlated shocks to both human and physical capital, and (ii) an application and damage specification to natural disaster risk, specifically tropical cyclones.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

of the empirical literature. For example, the model predicts both negative and partly persistent output impacts from cyclone *strikes* (e.g., Raddatz, 2007; Strobl, 2011; Hsiang and Jina, 2014; Elliott *et al.*, 2015) and ambiguous growth impacts of cyclone *risk*, mediated through channels such as relatively higher human capital accumulation (e.g., Skidmore and Toya, 2002). This section first presents the model and then proceeds to describe some of its key features below.

2.1. *Model Setup*

Each country j is inhabited by a representative household who can invest in human capital ($h_{j,t}$) and physical capital ($k_{j,t}$). Each period t, the economy faces a cyclone shock $\varepsilon_{j,t}$ that is independent and identically distributed, $\varepsilon_{j,t} \sim f_j(\varepsilon_j)$. The variable $\varepsilon_{j,t}$ here denotes cyclone intensity, and is later quantified to represent maximum sustained wind speeds per unit of land area in a given year. The dependence of its distribution on the climate is suppressed at this stage for notational simplicity. Cyclones can destroy both physical and human capital. Formally, the fractions of physical and human capital destroyed by a storm of intensity $\varepsilon_{j,t}$ are denoted by $\eta_j^k(\varepsilon_{j,t})$ and $\eta_j^h(\varepsilon_{j,t})$, respectively. The relationship between storm intensity and damage also varies across countries j.[21] The representative agent in country j chooses state-contingent plans for consumption $c_{j,t}$ and investments in human and physical capital ($x_{j,t}^h, x_{j,t}^k$) to maximise[22]

$$\begin{aligned} &\max E_{j,0} \sum_{t=0}^{\infty} \beta^t u(c_{j,t}) \\ \text{such that} \quad & c_{j,t} + x_{j,t}^k + x_{j,t}^h = k_{j,t} R_{j,t}^k + h_{j,t} R_{j,t}^h, \\ & k_{j,t+1} = (1 - \delta_k - \eta_j^k(\varepsilon_{j,t}))k_{j,t} + x_{j,t}^k, \\ & h_{j,t+1} = (1 - \delta_h - \eta_j^h(\varepsilon_{j,t}))h_{j,t} + x_{j,t}^h, \\ & k_{j,0}, h_{j,0} \text{ given and } k_{j,t} \geqslant 0, h_{j,t} \geqslant 0. \end{aligned} \tag{2}$$

Here, $R_{j,t}^k$ and $R_{j,t}^h$ denote returns to physical and human capital and δ_m denotes baseline depreciation of asset m. Conceptually, baseline human capital depreciation reflects factors such as skill depreciation (due to, e.g., unmodelled unemployment as in Dinerstein *et al.*, 2022) and baseline mortality (which necessitates new investment in human capital of the young even with steady population levels). Specification (2) assumes that both physical and human capital can be quickly rebuilt after a storm event. Though necessary for tractability, this assumption is likely to bias our estimates of disaster impacts downwards, as discussed below.

Aggregate production by the representative firm rents households' factors $K_{j,t} \equiv k_{j,t} L_j$ and $H_{j,t} \equiv h_{j,t} L_j$ in competitive national markets, where L_j denotes the country's population. The representative firm maximises intratemporal profits:

$$\max_{K_{j,t}, H_{j,t}} A_{j,t}(\varepsilon_{j,t}) K_{j,t}^{\alpha} H_{j,t}^{1-\alpha} - R_{j,t}^k K_{j,t} - R_{j,t}^h H_{j,t}. \tag{3}$$

[21] For now, we suppress the time subscripts on the damage functions $\eta_j^h(\cdot), \eta_j^k(\cdot)$ as the model treats these as constant within the current steady state. In comparing present and future steady states, however, we later allow for the possibility that damage functions change along with cyclone risks.

[22] As is standard in this type of model, there is no non-negativity constraint on human capital investment (see, e.g., Krebs, 2003b; Krebs *et al.*, 2015). In theory, such disinvestment could represent, for example, agents taking on more physically demanding labour that diminishes their health (e.g., mining). In practice, we do not find disinvestment in our quantitative analysis. That is, a non-negativity constraint would likely not be binding in our quantitative model. Even after a simulated category 5 hurricane, human capital investment remains non-negative for all countries.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Here $A_{j,t}(\varepsilon_{j,t})$ denotes TFP, which is also a function of storm realisations $\varepsilon_{j,t}$.

2.1.1. *Equilibrium growth and welfare*

We can now derive expressions for both expected long-run growth and realised annual growth in country j. First, let $\widetilde{k}_{j,t} \equiv k_{j,t}/h_{j,t}$ denote the household's ratio of physical to human capital. In equilibrium, by market clearing, the household's and the aggregate physical-human capital ratios are equal, $\widetilde{k}_{j,t} = \widetilde{K}_{j,t} \equiv K_{j,t}/H_{j,t}$, implying that factor returns consistent with profit-maximisation are

$$R^k_{j,t} = (\alpha)A_{j,t}(\varepsilon_{j,t}) \cdot (\widetilde{k}_{j,t})^{\alpha-1}, \qquad R^h_{j,t} = (1-\alpha)A_{j,t}(\varepsilon_{j,t}) \cdot (\widetilde{k}_{j,t})^{\alpha}. \tag{4}$$

Next, let the household's wealth at time t be defined by the sum of its physical and human capital: $w_{j,t} \equiv k_{j,t} + h_{j,t}$. The household's realised rate of return on this wealth in period t is given by the weighted sum of the net returns on physical and human capital:

$$\begin{aligned} r_j(\widetilde{k_{j,t}}, \varepsilon_{j,t}) \equiv\ & \omega_k(\widetilde{k_{j,t}})[R^k_{j,t}(\widetilde{k}_{j,t}, \varepsilon_{j,t}) - \delta_k - \eta^k_j(\varepsilon_{j,t})] \\ & + [1 - \omega_k(\widetilde{k_{j,t}})][R^h_{j,t}(\widetilde{k}_{j,t}, \varepsilon_{j,t}) - \delta_h - \eta^h_j(\varepsilon_{j,t})]. \end{aligned} \tag{5}$$

Here $\omega_k(\widetilde{k_{j,t}}) \equiv \widetilde{k}_{j,t}/(1+\widetilde{k}_{j,t})$ denotes the share of the household's wealth invested in physical capital. Intuitively, (5) already showcases multiple channels through which cyclones may affect household incomes. On the one hand, we see that the immediate destruction of both physical and human capital (via $\eta^k_j(\varepsilon_{j,t})$ and $\eta^h_j(\varepsilon_{j,t})$) directly leads to a decline in the household's returns on its assets, ceteris paribus. Negative impacts of storm realisations on TFP (via $A_{j,t}(\varepsilon_{j,t})$ in (4)) further reduce factor incomes, ceteris paribus. On the other hand, the potential for indirect or long-term impacts is also apparent. If households respond to cyclone risk by shifting their investments then this change in the physical-human capital ratio $\widetilde{k}_{j,t}$ could alter both the equilibrium factor returns (via (4)) and the relative importance of the asset destruction impacts (via $\omega_k(\widetilde{k_{j,t}})$). For example, if storms pose a bigger risk to physical than human capital, households may respond to storm risk by shifting their investments towards human capital, lowering their direct vulnerability, but also decreasing the returns to human capital on the margin. Our quantitative model predicts this adjustment to occur (see the Online Appendix).

A final channel through which cyclone risk can affect growth in the model is through savings rates. Let $\widetilde{s}_{j,t} \equiv 1 - c_{j,t}/\{w_{j,t}[1 + r_j(\widetilde{k}_{j,t}, \varepsilon_{j,t})]\}$ denote the household's savings-out-of-wealth ratio. From now on we assume that preferences follow a constant-elasticity-of-substitution form:

$$u(c_{j,t}) = \begin{cases} \dfrac{c_{j,t}^{1-\gamma}}{1-\gamma} & \text{if } \gamma \neq 1, \\ \log(c_{j,t}) & \text{if } \gamma = 1. \end{cases}$$

Following Krebs (2003a), it is straightforward to show (see the Online Appendix) that the capital ratio $\widetilde{k}_j$ and the savings rate $\widetilde{s}_j$ that solve the household's problem in stationary equilibrium (where $\widetilde{k}_{j,t} = \widetilde{k}_j$ and $\widetilde{s}_{j,t} = \widetilde{s}_j$) are determined by the two equations

$$\widetilde{s}_j = (\beta E_j[\{1 + r_j(\widetilde{k}'_j, \varepsilon'_j)\}^{1-\gamma}])^{1/\gamma}, \tag{6}$$

$$0 = \beta E_j\left(\frac{[R^k_j(\widetilde{k}_j, \varepsilon'_j) - \delta^k_j - \eta^k_j(\varepsilon_j\prime)] - [R^h_j(\widetilde{k}_j, \varepsilon'_j) - \delta^h_j - \eta^h_j(\varepsilon'_j)]}{[1 + r_j(\widetilde{k}'_j, \varepsilon'_j)]^{\gamma}}\right). \tag{7}$$

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Intuitively, optimal savings $\widetilde{s}_j$ follow from the household's Euler equation, whereas (7) expresses an optimality condition for $\widetilde{k}_j$ based on the risk-adjusted expected returns to human and physical capital. Intuitively, households face an optimal portfolio choice problem with two risky assets, so that their optimal investment allocation depends on expected returns as well as the variances and co-variances of the returns on each type of capital (see, e.g., Campbell and Viceira, 2002). Importantly, (7) does not imply that the expected returns to physical and human capital will be equated in equilibrium, but allows for the riskier asset to command a risk premium.

We can now characterise long-run or *average growth* as equal to (see the Online Appendix)

$$\overline{g}_j \equiv E\left[\frac{c_j'}{c_j}\right] = (\widetilde{s}_j)\{1 + E_j[r_j(\widetilde{k}_j', \varepsilon_j')]\}. \tag{8}$$

Realised year-to-year growth $g_{j,t}$, in turn, is given by

$$g_{j,t} = \frac{c_{j,t}}{c_{j,t-1}} = (\widetilde{s}_j)[1 + r_j(\widetilde{k}_{j,t}, \varepsilon_{j,t})]. \tag{9}$$

Equations (6)–(9) illustrate how cyclone risk can alter growth through its potential impact on savings rates. For example, if households respond to higher cyclone risk with increased (precautionary) savings $\widetilde{s}_j$, average growth (8) may increase, ceteris paribus. Our quantitative model predicts such increases in savings rates with increasing cyclone risk (see the Online Appendix). Cyclone strikes, however, unambiguously reduce returns and thus year-to-year growth (9). The structure of the model also implies that some of these losses will remain over time. This persistence is due to the fact that the model features constant returns to scale in reproducible factors, and as neither the optimal capital ratio nor the household's savings rate (6)–(7) depend on wealth levels (see also Online Appendix Figures A5 and A6 for a quantitative illustration). Though common in stochastic growth models of disasters and climate risk (e.g., Pindyck and Wang, 2013; Bretschger and Vinogradova, 2018; Van den Bremer and Van der Ploeg, 2021), there are well-known caveats associated with this modelling approach, such as its shortcomings in matching empirical moments such as on cross-country growth convergence (Mankiw *et al.*, 1992). While alternative models may thus better explain other aspects of cross-country growth, for our objective of capturing the marginal effects of cyclone risk changes, our model arguably provides a natural starting point in line with related literature. Another limitation of our approach is that we do not account for frictions and delays that may prevent economies from quickly rebuilding their physical and human capital stocks after disasters. Other scholars have developed models that capture these frictions in detail, most prominently Hallegatte *et al.* (2007). Importantly, we note that the primary focus of our analysis is on the long-run impacts of cyclone risk changes. In order to quantify the costs of individual cyclone strikes to the economy accurately, our analysis also considers a specification based on cumulative TFP impacts over a five-year period, as described in Section 3 below. To the extent that these cumulative impacts account for the costs imposed by some of the omitted frictions, they are thus still indirectly included in our analysis. To the extent that delays in the rebuilding of human capital stocks in particular are still not captured, however, we ultimately consider our estimates a likely lower bound on cyclone costs.[23]

[23] Several empirical studies have documented very long-run and even intergenerational impacts of disasters (e.g., Caruso and Miller, 2015; Kousky, 2016). Our omission of rebuilding delays also implies that we may underestimate cumulative impacts in cases where cyclones repeatedly hit countries before they have time to rebuild their capital stock.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Finally, we highlight our measure of welfare. Following the same approach as Krebs (2003a), one can express the representative agent's expected lifetime utility $U_{0,j}$ as

$$\begin{aligned} U_{0,j} &\equiv E_{0,j} \sum_{t=0}^{\infty} \beta^t \frac{c_{j,t}^{1-\gamma}}{1-\gamma} \\ &= \sum_{t=0}^{\infty} \beta^t \frac{E_{0,j}[\{c_{0,j} g_{j,t}(\widetilde{k}_{j,t}, \varepsilon_{j,t})^t\}^{1-\gamma}]}{1-\gamma} \\ &= \frac{c_{0,j}^{1-\gamma}}{(1-\gamma)\{1-\beta E_{0,j}[\overline{g}_j(\widetilde{k}_j, \varepsilon_j)^{1-\gamma}]\}}. \end{aligned} \tag{10}$$

Here the second line follows from (9) and from the assumption that cyclone strikes are independently distributed. The third line follows from the fact that initial consumption $c_0 = (1-\widetilde{s})(1+r(\widetilde{k}_0, \varepsilon_0))w_0$ is pre-determined as initial assets h_0, k_0 and initial storm realisation ε_0 are given and as the savings rate (6) is constant in stationary equilibrium. We use (10) to evaluate the welfare impacts of cyclone risk in each country, specifically by computing the balanced growth path consumption equivalent, that is, the percentage by which consumption each period would have to be increased in order to make households indifferent to a cyclone risk change.

3. Model Quantification

This sections describes the quantification of our stochastic endogenous growth model for each country in the data. Sections 3.1 and 3.2 present the empirical estimation of direct cyclone impacts on TFP and the destruction of physical and human capital, respectively. Section 3.3 describes our quantification of the cyclone intensity probability distribution for each country under current and future climates. Finally, Section 3.4 presents the remaining calibration.

3.1. *Total Factor Productivity Impacts*

Though the empirical literature frequently focuses on GDP per capita growth as an outcome variable, these impact estimates are difficult to incorporate directly into macroeconomic models as GDP growth is typically endogenised. In contrast, climate impacts upon structural model *parameters* are straightforward to interpret and utilise. We thus begin by conducting a standard growth accounting exercise that decomposes cyclone output growth impacts into productivity versus factor input changes. The appropriate empirical specification will generally depend on the structure of the climate-economy model for which the estimates are intended. For the aggregate production structure in our framework (3), taking logs on both sides yields

$$\ln(A_{j,t}) = \ln(Y_{j,t}) - \alpha_{j,t} \ln(K_{j,t}) - (1-\alpha_{j,t})[\ln(hc_{j,t}) + \ln(L_{j,t}^{Pop})]. \tag{11}$$

We use Penn World Tables data on GDP, capital stocks, populations and human capital per worker to back out TFP $A_{j,t}$ for each country-year pair. The human capital variable maps (11) into the data following standard approaches (e.g., Hall and Jones, 1999) that specify human capital-augmented labour $H_{j,t}$ as the product of the number of workers $L_{j,t}$ and human capital per worker $hc_{j,t}$. The latter, in turn, is provided by PWT based on schooling data and returns to education estimates (Inklaar and Timmer, 2013). As our model features inelastic labour supply,

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Table 2. *TFP Impacts.*

	(1)	(2)	(3)	(4)	(5)
Max. Wind/sqkm$_{j,t}$	−1.512*	−1.693*	−1.993*	−2.118*	−2.071*
	(0.880)	(0.973)	(1.144)	(1.241)	(1.178)
Max. Wind/sqkm$_{j,t-1}$		−1.590**	−1.885**	−2.031*	−2.128*
		(0.750)	(0.913)	(1.040)	(1.108)
Max. Wind/sqkm$_{j,t-2}$			−1.932*	−2.072*	−2.164*
			(1.035)	(1.161)	(1.234)
Max. Wind/sqkm$_{j,t-3}$				−1.718	−1.837
				(1.165)	(1.252)
Max. Wind/sqkm$_{j,t-4}$					−1.487*
					(0.892)
Observations	6,161	6,033	5,905	5,777	5,649
Clusters	144	144	144	144	144
Adjusted R^2	0.642	0.639	0.635	0.631	0.625
AIC	−8447	−8386	−8327	−8255	−8160
BIC(n = #Clusters)	−8316	−8255	−8197	−8125	−8029

Notes: This table presents the regression of the natural log of countries' TFP on a constant, country and year fixed effects, country-specific linear time trends and the maximum wind speed per square kilometre. SEs are heteroscedasticity robust and clustered at the country level. ** $p < .05$, * $p < .1$.

we also use $L_{j,t}^{Pop}$ as a measure of workers. Following Gollin (2002) we assume common labour shares across countries and set $1 - \alpha_{jt} = 0.67$ for all j, t.

Our preferred specification de-trends each TFP series log-linearly through the inclusion of country-specific time trends $(\gamma_j \cdot t)$ and year fixed effects δ_t in an estimating equation that follows the standard panel approach (analogous to (1) but for TFP):

$$\ln(A_{j,t}) = \gamma_j + \delta_t + (\theta_j \cdot t) + \sum_{l=0}^{L} \beta_{1+l}^{A} \varepsilon_{j,t-l} + \epsilon_{j,t}.$$

Here the γ_j denote country fixed effects and the $\varepsilon_{j,t-l}$ are cyclone intensity measures up to lag L. SEs $\epsilon_{j,t}$ are heteroscedasticity robust and clustered at the country level. We consider a range of values of L. Table 2 presents results for cyclone intensity measured by maximum wind speed per square kilometre.

The results indicate significant negative impacts of cyclone strikes on TFP. While the Bayesian information criterion (BIC) and Akaike information criterion (AIC) are minimised for the contemporaneous impact specifications, we also find negative and at least marginally precisely estimated TFP impacts persisting up to around five years. The Online Appendix shows results for additional lags, which reduce the estimates' precision, but leave the magnitudes similar. The Online Appendix also presents results for specifications that (i) de-trend TFP through Hodrick–Prescott filtering, which leads to broadly similar results, and (ii) use dissipated energy as the cyclone intensity measure, which yields somewhat noisier estimates. Overall, the empirical results thus confirm that one of the channels through which cyclone strikes affect realised growth is by lowering TFP.[24]

The broader literature reveals a number of mechanisms through which TFP effects may occur. First, cyclones can cause extensive power outages; Allcott *et al.* (2016) showed how power outages map into reductions in measured productivity in a standard production framework.

[24] Loayza *et al.* (2012) considered a productivity impact channel for disasters by including capital investment rates in several output impact regressions, but did not estimate a structural damage function for TFP impacts.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Second, our framework assumes homogeneous capital stocks. In reality, production requires the combination of heterogeneous and often complementary capital inputs that cannot readily be substituted for each other, such as a road leading to a factory. Hallegatte and Vogt-Schilb (2019) showcased several mechanisms through which capital heterogeneity can lead to output losses conditional on capital levels in the aftermath of a disaster. Intuitively, storms cause not only a reduction in capital levels, but also a misallocation of capital across categories (e.g., cars versus roads).[25] General production input-output linkages and complementarities have also been linked to reductions in output after disasters even for firms not damaged by storms (that is, conditional on capital and labour); see, e.g., Boehm *et al.* (2019). Finally, disasters may displace labour through, e.g., evacuations or transportation disruptions. Such labour displacement can force temporary closures of businesses that were physically unaffected by an event (e.g., Tierney, 1997). In our production framework, a loss of output conditional on capital, labour and human capital would show up as a reduction in TFP. One implication of each of these mechanisms is that TFP should recover as repair and reconstruction activities restore critical infrastructure. In line with this prediction, our results do indeed suggest that TFP impacts dissipate over time, as we fail to detect significant effects after four to five years (see the Online Appendix).

We formally map the Table 2 results into model (3) via the damage function

$$A_{j,t}(\varepsilon_{j,t}) = \overline{A}_{j,t}(1 - \eta^A \cdot \varepsilon_{j,t}), \qquad \eta^A = \widehat{\beta_1^A},$$

where we use $\widehat{\beta_1^A}$ from column (5) of Table 2 as a benchmark estimate. For robustness, we also consider a cumulative five-year impact specification ($\eta^A(\varepsilon_{j,t}) = (\widehat{\beta_1^A} + \widehat{\beta_2^A} + \cdots + \widehat{\beta_5^A})\varepsilon_{j,t}$) based on column (5) of Table 2.

3.2. *Physical and Human Capital Losses*

While there is limited literature guidance for the estimation of cyclone TFP impacts, numerous studies have quantified cyclone destruction of property and human life as a function of storm characteristics. Following these studies (e.g., Kahn, 2005; Nordhaus, 2010b; Schumacher and Strobl, 2011; Hsiang and Narita, 2012), we specify damage as a polynomial function of cyclone intensity $\varepsilon_{j,t}$ (measured by maximum wind speed per square kilometre country area):

$$\begin{aligned} \eta^k_{j,t}(\varepsilon_{j,t}) &\equiv \frac{\text{PropertyDamage}_{j,t}}{K_{j,t}} = \xi^k_{1j,t}(\varepsilon_{j,t})^{\xi^k_{2j,t}}, \\ \eta^h_{j,t}(\varepsilon_{j,t}) &\equiv \frac{\text{Fatalities}_{j,t}}{L_{j,t}} = \xi^h_{1j,t}(\varepsilon_{j,t})^{\xi^h_{2,j,t}}. \end{aligned} \tag{12}$$

We note that the focus on fatalities as the only measure of human capital impacts is due to data limitations and expected to bias our cyclone impact estimates downward as it misses disruptions in education, morbidity and mental health effects of storms.

Our estimation allows the damage functions (12) to vary across countries and time, in line with prior literature. That is, letting $\mathbf{x}'_{j,t}$ denote a vector of country covariates (e.g., population share in low-lying coastal areas), we allow both the multipliers and exponents in (12) to vary with $\mathbf{x}'_{j,t}$

[25] To the extent that TFP impacts are due to capital misallocation, they may also be non-linear in parallel to capital losses. We leave further exploration of TFP impact channels and specifications as an important avenue for future work.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Table 3. *Physical and Human Capital Losses.*

Dependent variable:	ln(PropertyDamage$_{j,t}/K_{j,t}$)				ln(Fatalities$_{j,t}/L_{j,t}$)	
	(1)	(2)	(3)	(4)	(5)	(6)
ln(Max. Wind$_{j,t}$/sqkm$_j$)	1.311**	4.729***	2.051***	2.211***	0.985***	2.082***
	(0.523)	(0.958)	(0.555)	(0.551)	(0.220)	(0.339)
ln(Max. Wind$_{j,t}$/sqkm$_j$) · ln(GDP p.c.)$_{j,t-1}$			−0.162**	−0.196***		−0.162***
			(0.063)	(0.063)		(0.037)
ln(Max. Wind$_{j,t}$/sqkm$_j$)·(Pct. below 5 m)$_{j,t}$				0.021***		0.011**
				(0.007)		(0.005)
ln(GDP pc)$_{j,t-1}$			−1.904***	−2.286***		−2.185***
			(0.627)	(0.627)		(0.371)
(Pct. below 5 m)$_{j,t}$				0.182***		0.080**
				(0.057)		(0.038)
Constant	3.245	45.614***	13.770**	15.853***	−4.971**	11.291***
	(4.704)	(10.933)	(5.456)	(5.403)	(1.987)	(3.363)
Country fixed effects	Yes	U.S. Only	No	No	Yes	No
Observations	352	29	352	352	472	471
Adjusted R^2	0.0431	0.404	0.236	0.250	0.0454	0.493

Notes: This table presents the regression of the natural log of fractions of capital stock destroyed (columns (1)–(4)) or population killed (columns (5)–(6)) on the natural log of the maximum wind speed normalised by country area, lagged GDP per capita levels and maximum wind interactions (columns (3), (4), (6)), the percentage of population living below 5 m elevation in levels and maximum wind interactions (columns (4), (6)) and country fixed effects (columns (1), (5)). Column (2) restricts the sample to US storms only. Damage data source is EMDAT. Heteroscedasticity-robust SEs are reported in parentheses (*** $p < .01$, ** $p < .05$).

by estimating[26]

$$\ln(\eta^m_{j,t}) = \mathbf{x}'_{j,t}\beta^m + \beta^m_\varepsilon \ln \varepsilon_{j,t} + (\ln \varepsilon_{j,t} \cdot \mathbf{x}_{j,t})'\gamma^m + \epsilon_{j,t}, \qquad m \in \{k, h\}. \tag{13}$$

Specification (13) implies that the direct losses in, e.g., physical capital that a country experiences from a cyclone of a given intensity $\varepsilon_{j,t}$ depend on its characteristics. Given (13) one can infer the coefficients in (12) via

$$\widehat{\xi^m_{1,j,t}} = e^{\mathbf{x}'_{j,t}\widehat{\beta^m}}, \qquad \widehat{\xi^m_{2,j,t}} = \widehat{\beta}_\varepsilon + \mathbf{x}'_{j,t}\gamma^m. \tag{14}$$

We consider three variables as potential covariates $\mathbf{x}'_{j,t}$: country fixed effects, lagged GDP per capita and the share of the population living below 5 m elevation in coastal areas.[27] Table 3 displays the results for our preferred cyclone intensity measure of $\varepsilon_{j,t}$ as maximum wind speed per square kilometre of country area. As expected, capital losses are increasing in wind speeds, albeit with heterogeneous steepness across countries. Column (1) adopts country fixed effects as damage covariates $\mathbf{x}'_{j,t}$. This specification allows countries to differ in baseline damage conditional on experiencing a cyclone, but with common curvature in wind speed. Given the empirical literature's finding that damage curves are considerably steeper in the United States (e.g., Nordhaus, 2010b; Strobl, 2011) than globally (Hsiang and Narita, 2012; Bakkensen and Mendelsohn, 2016), column (2) presents a US-only specification, which confirms this pattern.[28] Finally, columns (4) and (6) allow damage to vary with countries' levels of economic development and the low-lying coastal area population share. As expected, the results indicate that both physical and human capital losses are significantly larger in countries with larger population shares in low-lying coastal

[26] Since we use the same explanatory variables for physical capital and fatality regressions, a seemingly unrelated regression approach would not change the results.

[27] While there are many other potential determinants of countries' cyclone vulnerabilities, these are the covariates we would expect to have first-order relevance and for which one can obtain projections of future levels, as is required for us to consider changes in countries' future vulnerabilities.

[28] Quantitatively, the results may differ from studies normalising damage by GDP as we study damage as a fraction of countries' capital stocks, which are not equiproportional to GDP across countries.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

areas, and significantly mitigated in countries with higher economic development. Importantly, these results also indicate that countries' future vulnerabilities to cyclones may decrease as they develop economically. The calibrated model thus uses the results of column (4) to quantify capital damage for all countries except the United States, for which we use the results from column (2). For fatalities, the calibration uses the results from column (6).

Table 3 is estimated using EMDAT data on cyclone damage. For robustness, we repeat the specification using MunichRe data (see Online Appendix Table A6). On the one hand, the MunichRe data yield steeper wind speed curvature estimates in the fixed-effect specifications (e.g., US damage elasticity of 5.9 instead of 4.7). On the other hand, the specifications with interaction terms are comparatively attenuated.

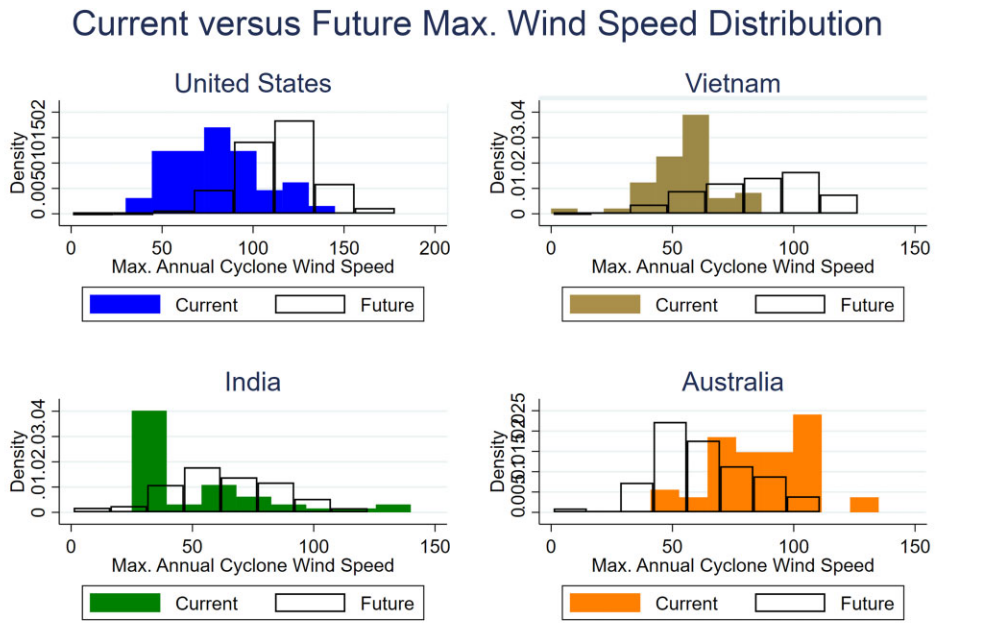
3.3. *Cyclone Risk Changes*

The empirical estimates presented thus far quantify the impacts of weather shocks $\varepsilon_{j,t}$. Linking these estimates to climate change requires a quantification of how the probability distribution of these shocks will change with global warming. That is, letting T_τ denote the global mean surface temperature in period τ, the quantification of our model requires estimates of cyclone probability density functions (PDFs) $f_j(\varepsilon|T_\tau)$. The availability of atmospheric science data to estimate such PDFs was previously limited, forcing some earlier literature to evaluate damage functions at the projected future *means* of cyclone intensity, effectively computing damage at expected intensity rather than expected damage (e.g., Narita *et al.*, 2009; see also the review by Ranson *et al.*, 2014). Of course, if damage functions (12) are convex, this approach risks underestimating expected future impacts. In this paper we gratefully take advantage of advances in climatological research from Kerry Emanuel and co-authors (Emanuel, 2008; Emanuel *et al.*, 2008; and as utilised by Mendelsohn *et al.*, 2012) to construct estimates of cyclone PDFs. Their work generates 68,000 simulated synthetic tropical cyclone tracks under each of the current (1980–2000) and future climates, specifically 2080–2100 under the IPCC's A1B emissions scenario and processed through four different general circulation models. The A1B scenario implies less warming than a business-as-usual scenario, with a best estimate of 2.8 °C by the end of the century over a 1980–99 baseline (IPCC, 2007). Our benchmark analysis focuses on results using NOAA's Geophysical Fluid Dynamics Laboratory model (Manabe *et al.*, 1991), yielding 17,000 simulated tracks. We also consider alternatives in the sensitivity analysis below. The synthetic cyclone tracks contain parallel information to the historical record, such as storm latitude, longitude and wind speeds at points along the track life. Recent literature has used synthetic tracks to inform both current cyclone risk assessments (Hallegatte, 2007; Elliott *et al.*, 2015) and projections of direct cyclone damage from climate change (Hallegatte, 2009; Mendelsohn *et al.*, 2012).

In order to estimate cyclone PDFs at the country-year level, we conduct Monte Carlo simulations based on current and future landfall frequencies and sampling from either the historical cyclone record (to estimate current risk) or from synthetic tracks (to estimate future risk), as described in the Online Appendix. Importantly, this process captures changes in expected future intensity driven both by changes in the *number* and *characteristics* of storms. For landfall frequencies, we adopt a Poisson distribution (Emanuel, 2013). For our preferred cyclone measure of maximum wind speeds, the literature has found Weibull distributions to provide the best fit (Johnson and Watson, 2007), which we consequently use to estimate cyclone PDFs for

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025



Current reflects IBTRACS data 1970–2015.
Future reflects 2090–2100 in the IPCC A1B scenario based on the synthetic cyclone tracks of Emanuel *et al.*(2008)

Fig. 1. *Example Comparisons of Current (1970–2015) versus Future (2100) Cyclone Risk Distributions under the IPCC A1B Scenario.*

each country.[29] To validate this approach, we compare the expected annual maximum wind speeds from the Weibull model against empirically observed means in the data. The Weibull model reproduces the data extremely well, with a correlation coefficient of 0.9985 (plotted in Online Appendix Figure A1).

In order to illustrate the potential impacts of climate change on cyclone risks, Figure 1 compares the current (from data) and projected future maximum wind speed distributions for four example countries. The simulations indicate highly heterogeneous impacts, with cyclone risk increases in some regions (e.g., the United States), but decreases in others (e.g., Australia). Countries are also predicted to experience changes in the variability of cyclone intensity (e.g., India). Figure 2 presents results for a broader set of countries, specifically by comparing current annual maximum wind speeds (x axis) against expected future annual maximum wind speeds (y axis). By comparing the location of each point (country) against the plotted 45° line, we again see that some countries are predicted to experience substantial increases in average risk, whereas others are predicted to see declines in average cyclone activity. The Online Appendix also presents analog versions to Figure 2 for maximum wind speed per square kilometre of country area, in line with the intensity measure used in the quantitative model.

[29] While 'fat tails' have been noted as a concern for some climate risks, cyclone wind speeds face a physical upper bound (Holand and Emannuel, 2011), and fitting even a log-normal distribution can imply 'meteorologically unrealistic' upper tail behaviour of excessive wind speeds (Johnson and Watson, 2007). Relatedly, Conte and Kelly (2016) found that cyclone damage in the United States follows a fat-tailed distribution due to the spatial distribution of properties, but that household-level damage and the wind speed distribution are thin tailed. We account for uniquely high US damage by utilising a separate capital depreciation elasticity.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

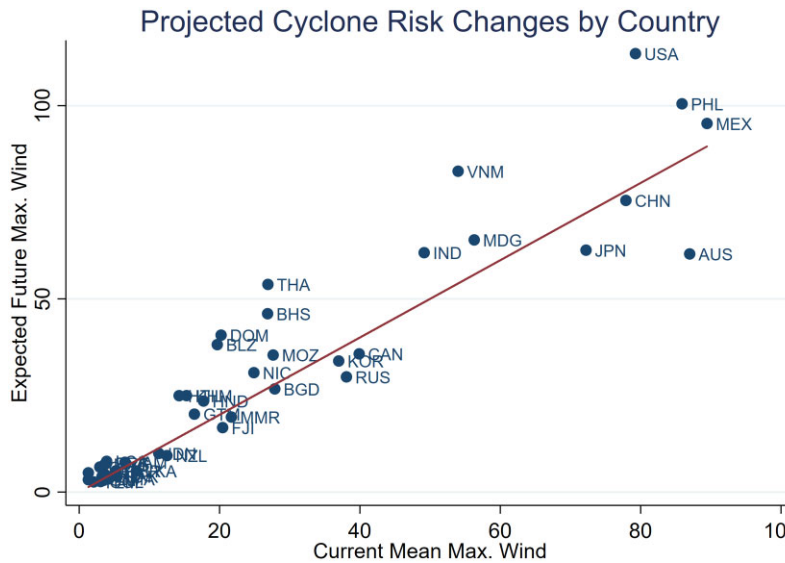


Fig. 2. *Current (1970–2015) Average Maximum Sustained Wind Speed (X Axis) versus Expected Future (2100) Mean Maximum Wind Speed (Y Axis) in the IPCC A1B Scenario based on Estimates from Emanuel (2008).*
Notes: Country names are provided in Online Appendix Table A7.

3.4. *Remaining Model Elements*

We quantify the remainder of the model for each country using a combination of data, estimation, matching of moments and external calibration. Table 4 summarises the calibration strategy. Initial capital stock levels ($K_{2014,j}$) are taken directly from the data (Penn World Tables), with a model base year of 2014. Human capital stocks are backed out via $H_{2014,j} = K_{2014,j}/\widetilde{k}_{2014,j}$ after solving the model for initial asset allocation ratios ($\widetilde{k}_{2014,j}$) via (7). Baseline (i.e., non-cyclone) depreciation rates (δ_k, δ_h) are calibrated in line with the literature,[30] as are the capital share (α), utility discount factor (β) and coefficient of relative risk aversion (γ). Initial productivity levels $\overline{A}_{2014,j}$ are chosen to match observed base year GDP per capita growth $g_{2014,j}$ (from the Penn World Tables) at base year cyclone realisations $\varepsilon_{2014,j}$ (from IBTrACS) in each country.[31] Our counterfactual analysis considers the impacts of changes in cyclone risk both in today's economy and at the end of the century. We set future productivity $\overline{A}_{2095,j}$ and baseline wealth levels $W_{2095,j}$ to match projected year 2095 GDP per capita levels and growth rates (sans cyclone risk changes) based on projections from the RICE model (Nordhaus, 2010a; 2011).[32] The corresponding physical and human capital stocks then follow after solving for the optimal asset allocation $\widetilde{k}_{2095}$

[30] For human capital, literature estimates vary. For example, Krebs *et al.* (2015) set δ_h equal to 0.0429 per year based on a target value for the ratio of human capital investment to gross domestic product. Manuelli and Seshadri (2014) jointly matched a set of moments with δ_h equal to 0.027. Some older literature adopts higher values: Jones *et al.* (1999) assumed δ_h equal to 0.075 and Krebs (2003a) adopted a value of 0.06. We thus select 0.04 as an intermediate value across this range.

[31] At this step we drop some countries from the sample that experienced negative growth in 2014, and in some cases severely so due to warfare or other non-modelled crises (e.g., Venezuela).

[32] That is, as the RICE model does not account directly for cyclone impacts, we calibrate our model so that it matches the RICE predictions in a scenario without cyclone risk changes.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Table 4. *Model Calibration.*

Country specific from data, estimation matched moments		
Item	Description	Source or value
$K_{2014,j}$	Capital stock	PWT data
$H_{2014,j}$	Human capital	Back out via $H_{2014,j} = K_{2014,j}/\widetilde{k}_{2014,j}$
$f_j(\varepsilon \mid T_\tau)$	Cyclone risk PDF	Estimated from IBTrACS cyclone records, synthetic tracks of Emanuel *et al.* (2008); see Section 3.3
$\eta_j^k(\varepsilon)$, $\eta_j^h(\varepsilon)$, $\eta_j^A(\varepsilon)$	Damage functions	Estimated from global macro panel, damage data; see Sections 3.1, 3.2
$\overline{A}_{2014,j}$	TFP	Match initial GDP growth (PWT)
$\overline{A}_{2095,j}$	TFP	Match projected 2095 GDP per capita levels, subsequent growth rates absent cyclone risk changes from Nordhaus (2011)
$W_{2095,j}$	Wealth	
$K_{2095,j}$	Capital stock	Back out via $K_{2095,j} = W_{2095,j}(\widetilde{k}_{2095,j}/(1+\widetilde{k}_{2095,j}))$
$H_{2095,j}$	Human capital	Back out via $H_{2095,j} = W_{2095,j}(1/(1+\widetilde{k}_{2095,j}))$
Globally standard from the literature		
α	Capital share	0.33
δ_k	Baseline depreciation	10%/yr
δ_h	Baseline depreciation	4%/yr
γ	Risk aversion	2
β	Utility discount	0.975

via (7). Finally, for scenarios that evaluate damage functions (14) at future covariate levels, we use projections of low-lying coastal population shares in 2100 from CIESIN.

3.5. *Model Assessment*

Before proceeding to counterfactual simulations, we compare some of the model's predictions to empirical counterparts that were not targeted in the calibration. First we compare the model's projected output growth impacts of cyclone strikes with the reduced-form empirical evidence presented in Section 1. That is, we use the structural model to predict the contemporaneous output growth impacts of a hypothetical category 5 hurricane striking each country in the base year. We then derive an analogous prediction from the reduced-form empirical evidence in Table 1.[33] Figures 3 and 4 show scatter plots of the resulting structural versus empirical predictions among the full sample of countries and an enlarged view among countries with predicted growth impacts of less than 5 percentage points, respectively. Both figures show that the structural model predictions align well with the empirical reduced-form results.

As a second assessment, we compare the model's predictions for investment in physical capital to data on gross fixed capital formation in the model base year. While the model's predictions are quantitatively reasonable, ranging from 20.2% to 26.9%, they do not correlate well with variation across countries in the data. This is arguably not surprising as our model is designed to capture marginal changes in investment rates due to variation in storm risks and thus abstracts from broader determinants of savings and investment such as age dependency ratios, pension systems, fiscal policy, capital controls, etc. In order to ensure the robustness of our results to this potential shortcoming, we implement an alternative calibration that allows utility discount

[33] We specifically use the Table 1 results in column (6) for maximum wind speed per square kilometre and evaluate the predicted impacts at each country's size and base year GDP. The results using the more precisely estimated coefficients in column (4) of Table 1 instead are extremely similar and shown in the Online Appendix.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

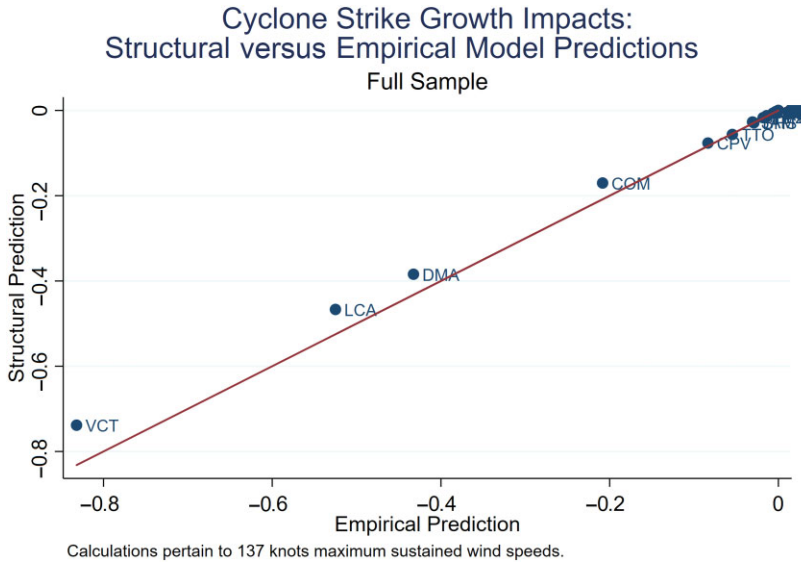


Fig. 3. *Comparison of Empirical versus Structural Model Predictions of Output Growth Impacts from a Category 5 Cyclone for Each Country (Dots) Along With the* 45° *Line.*

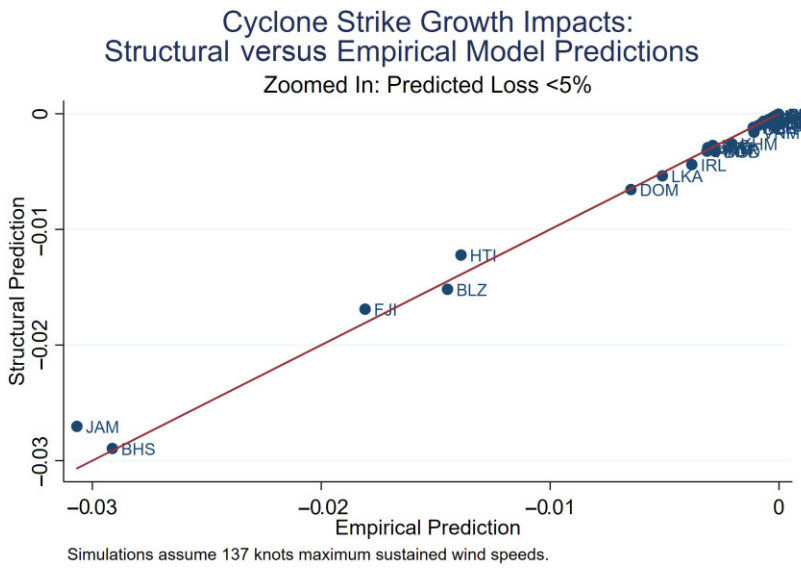


Fig. 4. *Enlarged Comparison of Empirical versus Structural Model Predictions of Output Growth Impacts from a Category 5 Cyclone for Each Country (Dots) Along With the* 45° *Line among Countries with Predicted Loss* $< 5\%$.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Table 5. *Model Scenario Summary.*

	Benchmark			Comparison		
Scenario	Economy	Vulnerability	Cyclone risk	Economy	Vulnerability	Cyclone risk
#1	Baseline	Baseline	Baseline	Baseline	Baseline	Zero
#2	Baseline	Baseline	Baseline	Baseline	Baseline	Future
#3	Future	Baseline	Baseline	Future	Baseline	Future
#4	Future	Future	Baseline	Future	Future	Future

factors β_j to vary across countries in order to match observed base year capital investment rates. The Online Appendix details this exercise. Importantly, the results are broadly similar to the benchmark model for welfare and virtually identical for the predicted growth impacts of cyclone risk.

As a final assessment, we also verify that the model predicts persistence in output losses from storms. While this is a qualitative feature of the model, we also showcase the quantitative predicted output dynamics following a hypothetical storm for several example countries in the Online Appendix. The results confirm that, while the contemporaneous negative growth impact is mitigated after the storm, some output losses are persistent, in line with the empirical evidence (e.g., Raddatz, 2007; Strobl, 2011; Hsiang and Jina, 2014; Elliott *et al.*, 2015).

4. Model Results

Our main results consider the following four scenarios, as summarised in Table 4. First, we evaluate the effects of present-day cyclone risk by comparing each country's observed baseline economy to a counterfactual one with cyclone risk set to zero.[34] Second, we consider the ceteris paribus effect of changing cyclone risks to future levels (2095–2105) consistent with our benchmark warming scenario. The predicted changes in cyclone risk for each country are displayed in Figure 2 in terms of maximum wind speed, and in Online Appendix Figures A2 and A3 in terms of maximum wind speed per square kilometre. This scenario leaves both the economy and cyclone vulnerability—specifically damage functions (see (12), (14))—at baseline levels. Third, we consider the impacts of cyclone risk *changes* in each country's projected future economy. That is, we adjust technology levels and capital stocks to be consistent with future GDP projections (2095) and evaluate the impacts of warming-induced cyclone risk changes in this setting. One comparison assumes that countries' vulnerabilities remain at baseline levels (specifically by maintaining damage functions (see (12), (14)) at their 2015 levels). The other assumes that cyclone vulnerability will continue to decline with growing income levels, as suggested in Table 3 (specifically by evaluating damage functions at 2095 GDP levels and low-lying population shares)[35]. Table 5 summarises the four different comparison scenarios we consider.

Figure 5 shows the estimated welfare impacts of both current and future cyclone risks on each country's baseline economy (scenarios 1 and 2). First, the results show that many countries are already suffering large welfare losses from present-day cyclone risk, with welfare losses of over 5% in some small island states. Second, global warming is projected to dramatically increase

[34] For comparability, we keep observed 2014 cyclone realisations, but set future risk expectations to zero.

[35] We assume that income growth cannot render the damage function coefficients negative by setting damage function coefficients to zero in cases where the projection would imply a negative value.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

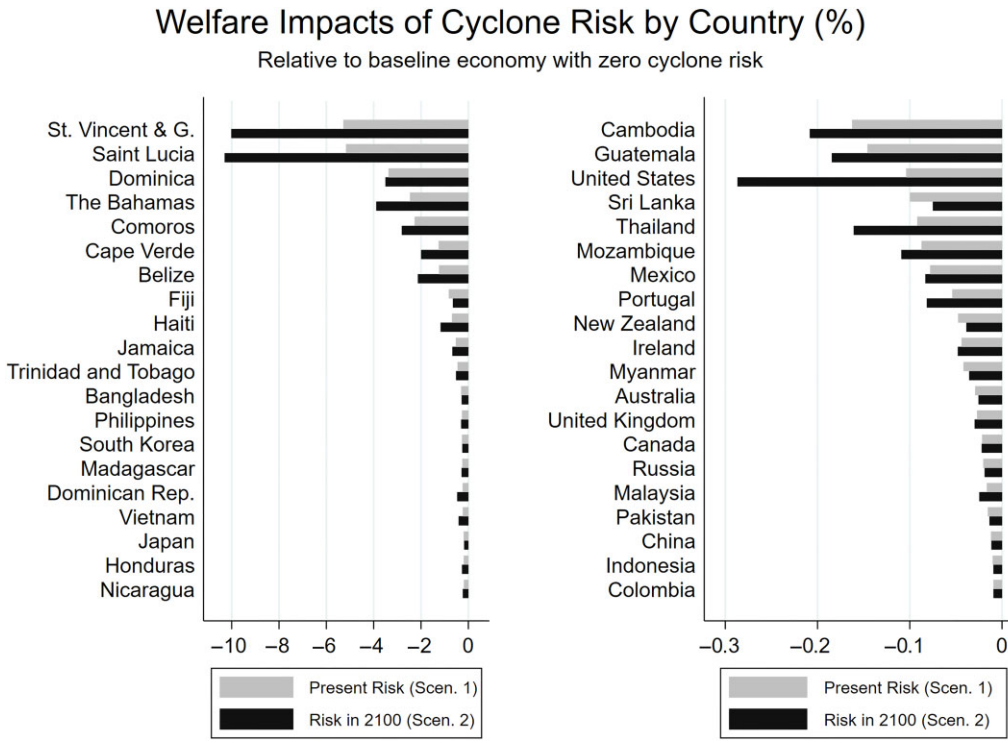


Fig. 5. *Welfare Impacts of Present (1970–2015, Scenario 1) and Future (2100, Scenario 2) Cyclone Risk on Each Country's Baseline Economy.*
Notes: Countries are ordered by present-day impacts.

these effects in many countries. Some of the most severe increases are projected to occur in countries that are already among the most vulnerable, such as Caribbean island states. Strikingly, the United States also stands out as having an almost three-fold projected increase in welfare costs from cyclones due to warming. While some countries are also projected to see declines in welfare costs, consistent with Figure 2, these generally appear smaller in magnitude (e.g., Australia). Figure 6 displays the corresponding growth rate impacts of current and future cyclone risks in each country's baseline economy (scenarios 1 and 2). Growth impacts broadly mirror projected welfare losses, but are smaller in magnitude. Interestingly, we also find that cyclone risk decreases growth in all countries, despite leading to increases in savings rates. This result is at odds with the cross-sectional empirical findings of Skidmore and Toya (2002) suggesting a positive correlation between cyclone risks and growth. This difference could be indicative of omitted variable bias in the cross-country regression, although we of course cannot rule out model misspecification.[36]

Our model enables us to distinguish growth impacts from the direct output effects (via capital destruction, fatalities and lower TFP) and indirect behavioural effects (via changes in savings rates and investment allocation) of cyclone risk. Figure 7 presents a comparison of the direct

[36] One could also argue that our model omits some potentially growth-enhancing consequences of cyclone risk, such as positive externalities from increased human capital accumulation, as in Lucas (1988).

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

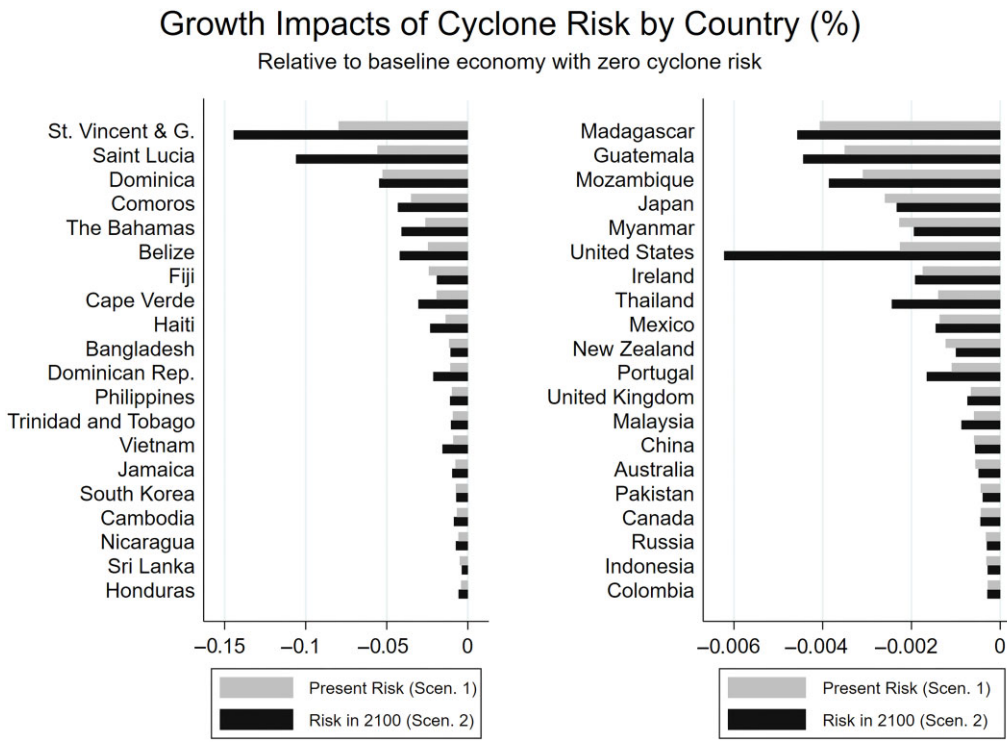


Fig. 6. *Growth Impacts of Present (1970–2015, Scenario 1) and Future (2100, Scenario 2) Cyclone Risk on Each Country's Baseline Economy.*
Notes: Countries are ordered by present-day impacts.

output and total (direct output + indirect behavioural) impacts of future cyclone risks on growth, relative to a baseline economy with zero cyclone risk as in scenario 2. The results indicate that the total impacts are usually smaller than direct output impacts. That is, behavioural responses can counteract the negative direct growth impacts of increasing cyclone risk by up to 30%–40% (e.g., in Belize, Vietnam, the United States).

Next, Figures 8 and 9 display projected welfare and growth impacts, respectively, of increasing cyclone risk from 2015 levels to 2100 levels in the projected future economy of each country (scenarios 3 and 4). Climate change is projected to have substantial negative welfare impacts through cyclone risk changes even in the wealthier economies of the future. Interestingly, the United States stands out among the 'top 10' of most negatively impacted countries, which are otherwise mostly poor and/or small island states. This result is driven by the combination of large increases in future hurricane risk in the United States (see, e.g., Figures 1 and 2) and the uniquely high vulnerability of the United States to cyclones of a given intensity (see, e.g., Table 3, column (1) versus (2)) in spite of its levels of income and experience (see also, e.g., Bakkensen and Mendelsohn, 2016). The results also suggest that some countries may benefit from future cyclone risk changes, in line with Figure 2 showing that some countries may see cyclone activity reductions as the climate warms. However, these benefits are quantitatively small, not consistent across climate models and subject to major caveats. Most importantly, our analysis does not

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

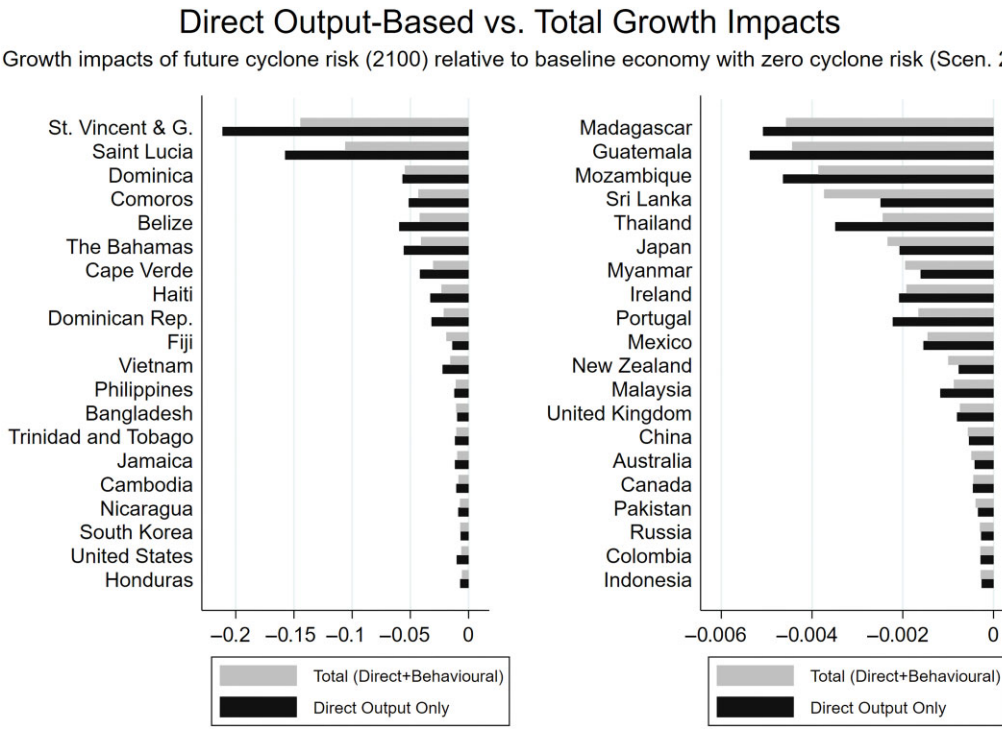


Fig. 7. *Direct Output-Based versus Total (Direct + Indirect Behavioural) Impacts of Future Cyclone Risk (2100) Relative to the Baseline Economy with Zero Cyclone Risk.*
Notes: Countries are ordered by total impacts.

account for sea level rise, which poses a major threat to some of the projected 'beneficiary' countries such as Fiji and Myanmar. Higher sea levels increase cyclone damage conditional on storm intensity. Whether the projected reductions in cyclone intensity would still translate into reductions in damage once higher sea levels were accounted for is thus highly questionable.

The results presented thus far incorporate only the contemporaneous one-year impact of cyclones on TFP. The empirical results in Section 4 suggest that TFP losses persist for several years, however. Figure 10 presents results from an alternative model based on the cumulative five-year TFP impacts of storms. The estimated welfare impacts are considerably larger than in the benchmark. For example, the welfare loss of future cyclone risk in Jamaica (scenario 2) rises from 0.75% to more than 2%.[37]

The Online Appendix presents additional model results for cyclone risk projections based on three alternative climate models (MIROC, CNRM and ECHAM). The results are mostly similar, although some differences arise. For example, some of the welfare benefits from cyclone risk reductions predicted by the benchmark model are not robust to consideration of other climate

[37] Some of the most vulnerable countries no longer have valid impact estimates in this specification. They end up as 'dismal cases' in that their expected damage can no longer be properly computed over the full integral of their Weibull wind distributions. We note that the five-year specification likely exaggerates expected damage as cumulative impacts are aggregated to occur within one model period.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

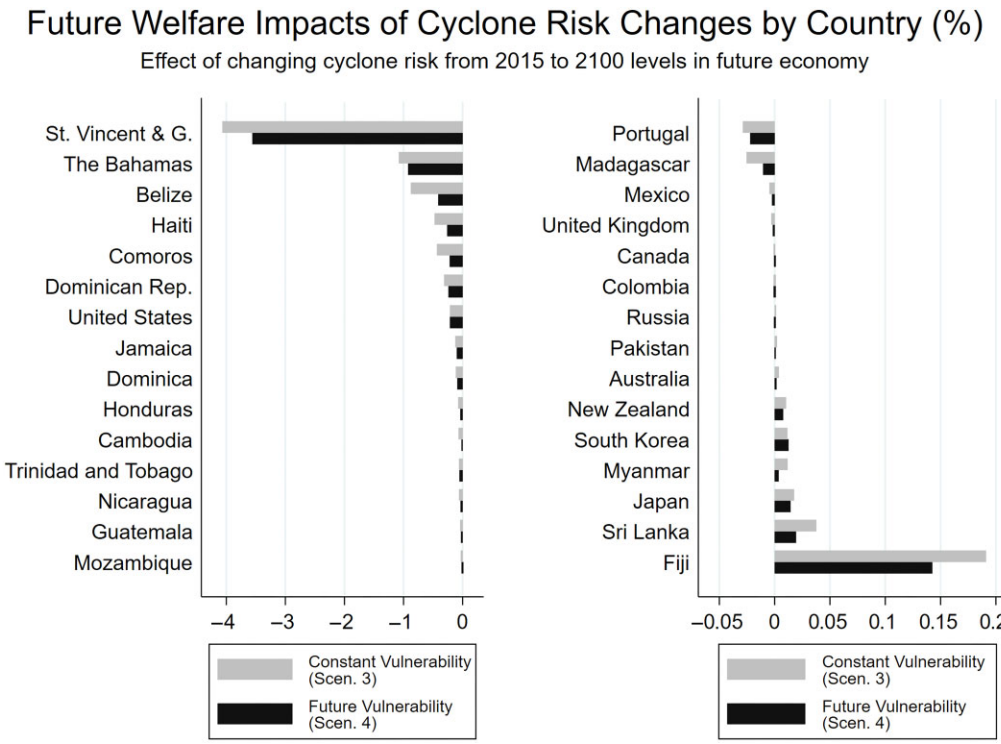


Fig. 8. *Welfare Impacts of Future Cyclone Risk Changes in 2100 with Constant (Scenario 3) or Future (Scenario 4) Damage Functions.*
Notes: Countries are ordered by impacts with constant vulnerability.

models. On the other hand, the other climate models also predict smaller future welfare costs in the United States.

The results presented thus far are disaggregated by country. Given that several cyclone-vulnerable countries are excluded from our analysis (due to, e.g., data limitations), we can only produce a *lower bound* on global impacts by computing the GDP-weighted global aggregate effects. The results indicate that the global aggregate costs of present-day cyclone risk (scenario 1) are equivalent to at least a 0.044% permanent consumption change.[38] The total costs of future cyclone risks (scenario 2) are at least 0.074% in balanced growth path consumption equivalent. To put these numbers in context, Barrage (2020) estimated total global welfare gains from optimised climate policy in the twenty-first century of 0.95% in balanced growth consumption equivalent.

Finally, we must caution that there are important drivers of future cyclone impacts that our model omits. For example, learning from increased cyclone exposure and investments in protective infrastructure will likely lower future vulnerability to storms below what our model predicts, ceteris paribus (see, e.g., Schumacher and Strobl, 2011; Hsiang and Narita, 2012 and Fried, 2022). On the other hand, however, our model also does not account for sea level rise, which will certainly increase future damage from storms (conditional on intensity). We also do not consider fiscal responses and impacts of storms, which can affect welfare costs through multiple

[38] Aggregating with population instead of GDP weights yields a slightly lower estimate of 0.037%.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

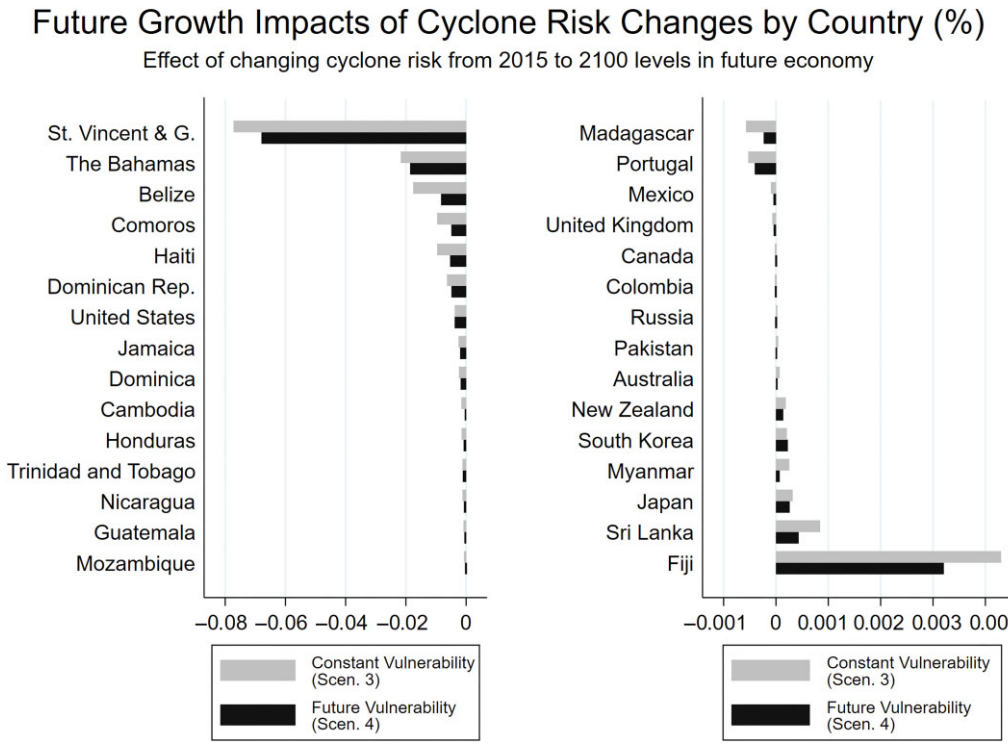


Fig. 9. *Growth Impacts of Future Cyclone Risk Changes in 2100 with Constant (Scenario 3) or Future (Scenario 4) Damage Functions.*
Notes: Countries are ordered by impacts with constant vulnerability.

channels (e.g,. Deryugina, 2017; Barrage, 2024; Phan and Schwartzman, 2024). Finally, the future locations of populations and economic activity have been shown to be critically important in determining climate change impacts (e.g., Desmet *et al.*, 2021). While our framework attempts to capture some of these dynamics based on exogenous predictions of future population shares in low-lying coastal areas, recent work has also demonstrated the potential for endogenous feedbacks between population locations, economic growth and exposure to storm damage (Hallegatte, 2017). Joint consideration of these dynamics in quantitative models of climatic risks is thus clearly an important area for future work.

5. Conclusion

This paper proposes a novel empirical-structural approach to analyse the macroeconomic consequences of climate change with a focus on tropical cyclones. We first empirically and conceptually review competing approaches to quantifying cyclone impacts on growth. We highlight that differences in reduced-form findings, driven in part by empirical choices, are maintained using a comprehensive dataset. Importantly, theory also tells us that even perfectly identified reduced-form regressions of growth on cyclone shocks or risk are individually insufficient to characterise

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

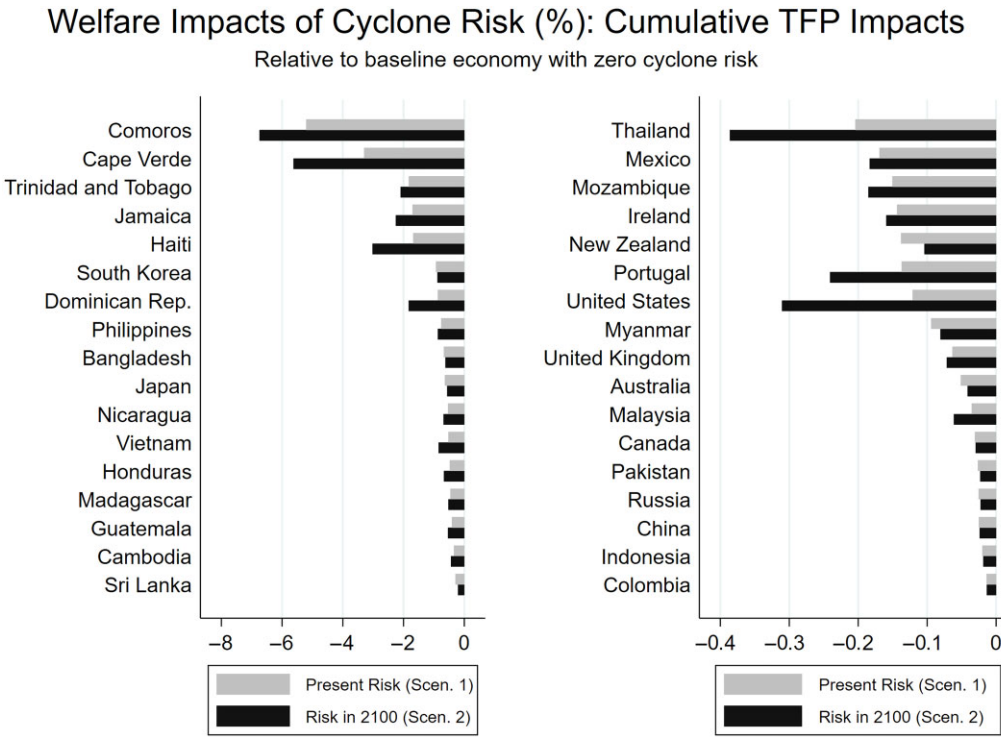


Fig. 10. *Welfare Impacts of Present-Day (Scenario 1) and Year 2100 (Scenario 2) Cyclone Risks on Each Country's Baseline Economy with Cumulative (Five-Year) TFP Effects.*

the full welfare effects of future changes in cyclone risks given broader general equilibrium changes.

Second, we present our approach to estimating and modelling cyclone impacts designed to combine empirical evidence with the structure of a model to deliver such welfare cost estimates. We propose that empirical research focus on quantifying cyclone impacts on the *structural determinants of growth*, and not just growth itself, as the latter is typically endogenous in macroeconomic models. We then present a stochastic endogenous growth cyclone-climate-economy model that we quantify separately for forty cyclone-vulnerable nations. Important for policy, we find significant heterogeneity of projected climate change impacts, including large negative welfare costs of up to −5% in balanced growth consumption equivalent in some of the most vulnerable locations. The United States stands out among the most negatively impacted countries. The model results also demonstrate the importance of accounting for macroeconomic adjustments to cyclone risk, which are projected to lower average growth impacts by up to 30%–40%.

A broader implication of our approach is to showcase the potential benefits of making the weather versus climate distinction explicit in a climate-economy model. This distinction permits (i) direct incorporation of plausibly causally identified cyclone impact estimates, (ii) accounting for macroeconomic adaptation through endogenous adjustments in savings and investments, and (iii) computing welfare costs of future changes in climatic risks. As frontier advancements in stochastic climate-economy models are now able to account for multiple sources of uncertainty

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

at high frequency (e.g., Cai and Lontzek, 2019), extending a truly integrated assessment model to explicit consideration of weather impacts may thus be an interesting area for future work and facilitate linkages to the empirical literature. Similarly, while our quantitative results are subject to numerous limitations ranging from our abstractions of advancements in wind-field modelling (Strobl, 2011; Hsiang and Narita, 2012) to distinguishing productive and adaptation capital (e.g., Fried, 2022), our proposed method seeks to complement these empirical and modelling advancements so as to facilitate the integration of both frontiers and to improve our understanding of the social costs of climate change.

University of Oregon, USA
ETH Zürich, Switzerland

Additional Supporting Information may be found in the online version of this article:

Online Appendix
Replication Package

References

Aiyagari, S. Rao. (1994). 'Uninsured idiosyncratic risk and aggregate saving', *The Quarterly Journal of Economics*, vol. 109(3), pp. 659–84.
Akao, Ken-Ichi and Hiroaki Sakamoto. (2013). 'A theory of disasters and long-run growth', Discussion papers 13061, Research Institute of Economy.
Allcott, Hunt, Allan Collard-Wexler, and Stephen D. O'Connell. (2016). 'How do electricity shortages affect industry? Evidence from India', *American Economic Review*, vol. 106(3), pp. 587–624.
Atkinson, Gary D. and Charles R. Holliday. (1977). 'Tropical cyclone minimum sea level pressure/maximum sustained wind relationship for the western North Pacific', *Monthly Weather Review*, vol. 105(4), pp. 421–7.
Auffhammer, Maximilian. (2018). 'Quantifying economic damages from climate change', *Journal of Economic Perspectives*, vol. 32(4), pp. 33–52.
Bakkensen, Laura A. and Robert O. Mendelsohn. (2016). 'Risk and adaptation: Evidence from global hurricane damages and fatalities', *Journal of the Association of Environmental and Resource Economists*, vol. 3(3), pp. 555–87.
Bakkensen, Laura and Lint Barrage. (2016). 'Do disasters affect growth? A macro model-based perspective on the empirical debate', Working Paper 2016-9, Brown University.
Bansal, Ravi and Marcelo Ochoa. (2011). 'Temperature, aggregate risk, and expected returns', Working Paper 17575, National Bureau of Economic Research.
Barrage, Lint. (2020). 'Optimal dynamic carbon taxes in a climate-economy model with distortionary fiscal policy', *The Review of Economic Studies*, vol. 87(1), pp. 1–39.
Barrage, Lint. (2024). 'The fiscal costs of climate change in the United States', Working Paper 23/380, CER-ETH.
Barro, Robert J. (2006). 'Rare disasters and asset markets in the twentieth century', *The Quarterly Journal of Economics*, vol. 121(3), pp. 823–66.
Barro, Robert J. and Jong Wha Lee. (2013). 'A new data set of educational attainment in the world, 1950–2010', *Journal of Development Economics*, vol. 104, pp. 184–98.
Bewley, Truman. (1977). 'The permanent income hypothesis: A theoretical formulation', *Journal of Economic Theory*, vol. 16(2), pp. 252–92.
Boehm, Christoph E., Aaron Flaaen, and Nitya Pandalai-Nayar. (2019). 'Input linkages and the transmission of shocks: Firm-level evidence from the 2011 Tōhoku earthquake', *Review of Economics and Statistics*, vol. 101(1), pp. 60–75.
Bovenberg, A. Lans and Sjak Smulders. (1995). 'Environmental quality and pollution-augmenting technological change in a two-sector endogenous growth model', *Journal of Public Economics*, vol. 57(3), pp. 369–91.
Bretschger, Lucas and Simone Valente. (2011). 'Climate change and uneven development', *The Scandinavian Journal of Economics*, vol. 113(4), pp. 825–45.
Bretschger, Lucas and Alexandra Vinogradova. (2018). 'Escaping Damocles' sword: Endogenous climate shocks in a growing economy', Working Paper 18/291, CER-ETH.
Burke, Marshall, Solomon M. Hsiang, and Edward Miguel. (2015). 'Global non-linear effect of temperature on economic production', *Nature*, vol. 527(7577), pp. 235–9.
Cai, Yongyang and Thomas S. Lontzek. (2019). 'The social cost of carbon with economic and climate risks', *Journal of Political Economy*, vol. 127(6), pp. 2684–734.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Campbell, John Y. and Luis M. Viceira. (2002). *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors*, Oxford: Oxford University Press.
Carleton, Tamma and Michael Greenstone. (2022). 'A guide to updating the US government's social cost of carbon', *Review of Environmental Economics and Policy*, vol. 16(2), pp. 196–218.
Caruso, Germán and Sebastian Miller. (2015). 'Long run effects and intergenerational transmission of natural disasters: A case study on the 1970 Ancash earthquake', *Journal of Development Economics*, vol. 117, pp. 134–50.
Cavallo, Eduardo, Sebastian Galiani, Ilan Noy, and Juan Pantano. (2013). 'Catastrophic natural disasters and economic growth', *Review of Economics and Statistics*, vol. 95(5), pp. 1549–61.
CIESIN. (2013). 'Low Elevation Coastal Zone (LECZ) Urban-Rural Population and Land Area Estimates, Version 2'. NASA Socioeconomic Data and Applications Center. https://doi.org/10.7927/H4MW2F2J (last accessed: January 2025).
Colacito, Ric, Bridget Hoffmann, and Toan Phan. (2019). 'Temperatures and growth: A panel analysis of the United States', *Journal of Money, Credit, and Banking*, vol. 51, 313–68.
Conte, Marc N. and David L. Kelly. (2016). 'An imperfect storm: Fat-tailed hurricane damages, insurance, and climate policy', Working paper, University of Miami.
CPI. (2015). 'Corruption Perceptions Index 2015', Transparency International, https://www.transparency.org/en/cpi/2015 (last accessed: January 2025).
Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken. (2012). 'Temperature shocks and economic growth: Evidence from the last half century', *American Economic Journal: Macroeconomics*, vol. 4(3), pp. 66–95.
Deryugina, Tatyana. (2017). 'The fiscal cost of hurricanes: Disaster aid versus social insurance', *American Economic Journal: Economic Policy*, vol. 9(3), pp. 168–98.
Desmet, Klaus, Robert E. Kopp, Scott A. Kulp, Dávid Krisztián Nagy, Michael Oppenheimer, Esteban Rossi-Hansberg, and Benjamin H. Strauss. (2021). 'Evaluating the economic cost of coastal flooding', *American Economic Journal: Macroeconomics*, vol. 13(2), pp. 444–86.
Devereux, Michael B. and Gregor W. Smith. (1994). 'International risk sharing and economic growth', *International Economic Review*, vol. 35(3), pp. 535–50.
Dietz, Simon and Nicholas Stern. (2015). 'Endogenous growth, convexity of damage and climate risk: How Nordhaus' framework supports deep cuts in carbon emissions', Economic Journal, vol. 125(583), pp. 574–620.
Dinerstein, Michael, Rigissa Megalokonomou, and Constantine Yannelis. (2022). 'Human capital depreciation and returns to experience', *American Economic Review*, vol. 112(11), pp. 3725–62.
Elliott, Robert, Jr., Eric Strobl, and Puyang Sun. (2015). 'The local impact of typhoons on economic activity in China: A view from outer space', *Journal of Urban Economics*, vol. 88, pp. 50–66.
Emanuel, Kerry A. (2013). 'Downscaling CMIP5 climate models shows increased tropical cyclone activity over the 21st century', *Proceedings of the National Academy of Sciences*, vol. 110(30), pp. 12219–24.
Emanuel, Kerry, Ragoth Sundararajan, and John Williams. (2008). 'Hurricanes and global warming: Results from downscaling IPCC AR4 simulations', *Bulletin of the American Meteorological Society*, vol. 89(3), pp. 347–68.
Emanuel, Kerry. (2008). 'The Hurricane—climate connection', *Bulletin of the American Meteorological Society*, vol. 89(5), pp. ES10–20.
Fankhauser, Samuel and Richard S.J. Tol. (2005). 'On climate change and economic growth', *Resource and Energy Economics*, vol. 27(1), pp. 1–17.
Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer. (2015). 'The next generation of the Penn World Table', *American Economic Review*, vol. 105(10), pp. 3150–82.
Fomby, Thomas, Yuki Ikeda, and Norman V. Loayza. (2013). 'The growth aftermath of natural disasters', *Journal of Applied Econometrics*, vol. 28(3), pp. 412–34.
Fried, Stephie. (2022). 'Seawalls and stilts: A quantitative macro study of climate adaptation', *The Review of Economic Studies*, vol. 89(6), pp. 3303–44.
Gollin, Douglas. (2002). 'Getting income shares right', *Journal of Political Economy*, vol. 110(2), pp. 458–74.
Greenstone, Michael. (2016). 'A new path forward for an empirical social cost of carbon', Presentation, https://sites.nationalacademies.org/cs/groups/dbassesite/documents/webpage/dbasse_172599.pdf (last accessed: July 2025).
Greenstone, Michael, Elizabeth Kopits, and Ann Wolverton. (2013). 'Developing a social cost of carbon for US regulatory analysis: A methodology and interpretation', *Review of Environmental Economics and Policy*, vol. 7, pp. 23–46.
Guha-Sapir, Debarati and Regina Below. (2002). 'The quality and accuracy of disaster data: A comparative analyses of three global datasets', Working paper, ProVention Consortium.
Guha-Sapir, Debarati, Regina Below, and Philippe Hoyois. (2016). 'EM-DAT: The CRED/OFDA International Disaster Database', Université Catholique de Louvain, Belgium .
Hall, Robert E. and Charles I. Jones. (1999). 'Why do some countries produce so much more output per worker than others?', *The Quarterly Journal of Economics*, vol. 114(1), pp. 83–116.
Hallegatte, Stephane and Adrien Vogt-Schilb. (2019). 'Are losses from natural disasters more than just asset losses?, in (Y. Okuyama and A. Rose, eds.), *Advances in Spatial and Economic Modeling of Disaster Impacts*, pp. 15–42, Cham, Switzerland: Springer.
Hallegatte, Stéphane, Jean-Charles Hourcade, and Patrice Dumas. (2007). 'Why economic dynamics matter in assessing climate change damages: Illustration on extreme events', *Ecological Economics*, vol. 62(2), pp. 330–40.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Hallegatte, Stéphane. (2007). 'The use of synthetic hurricane tracks in risk analysis and climate change damage assessment', *Journal of Applied Meteorology and Climatology*, vol. 46(11), pp. 1956–66.
Hallegatte, Stéphane. (2009). 'Roadmap to assess the economic cost of climate change with an application to hurricanes in the United States', in (J. Elsner and T. Jagger, eds.), *Hurricanes and Climate Change*, pp. 361–86, Boston, MA: Springer.
Hallegatte, Stephane. (2017). 'A normative exploration of the link between development, economic growth, and natural risk', *Economics of Disasters and Climate Change*, vol. 1(1), pp. 5–31.
Holand, Greg and Kerry Emannuel. (2011). 'Limits on hurricane intensity', Technical Report, https://emanuel.mit.edu/limits-hurricane-intensity (last accessed: June 2018).
Hsiang, Solomon M. and Amir S. Jina. (2014). 'The causal effect of environmental catastrophe on long-run economic growth: Evidence from 6,700 cyclones', Working Paper No. 20352, National Bureau of Economic Research.
Hsiang, Solomon M. and Amir S. Jina. (2015). 'Geography, depreciation, and growth', *American Economic Review*, vol. 105(5), pp. 252–6.
Hsiang, Solomon M. and Daiju Narita. (2012). 'Adaptation to cyclone risk: Evidence from the global cross-section', *Climate Change Economics*, vol. 03(02), 1250011.
Ikefuji, Masako and Ryo Horii. (2012). 'Natural disasters in a two-sector model of endogenous growth', *Journal of Public Economics*, vol. 96(9–10), pp. 784–96.
Inklaar, Robert and Marcel P. Timmer. (2013). 'Capital, labor and TFP in PWT8. 0', https://www.rug.nl/ggdc/docs/capital_labor_and_tfp_in_pwt80.pdf (last accessed: July 2025).
IPCC. (2007). *Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change*, pp. 1–996, Cambridge: Cambridge University Press.
Johnson, Mark E. and Charles C. Watson, Jr. (2007). 'Fitting statistical distributions to data in hurricane modeling', *American Journal of Mathematical and Management Sciences*, vol. 27(3–4), pp. 479–98.
Jones, Larry E., Rodolfo Manuelli, and Ennio Stacchetti. (1999). 'Technology (and policy) shocks in models of endogenous growth', Working Paper 7063, National Bureau of Economic Research.
Kahn, Matthew E. (2005). 'The death toll from natural disasters: The role of income, geography, and institutions', *Review of Economics and Statistics*, vol. 87(2), pp. 271–84.
Kalkuhl, Matthias and Leonie Wenz. (2020). 'The impact of climate conditions on economic production. Evidence from a global panel of regions', *Journal of Environmental Economics and Management*, vol. 103, 102360.
Knapp, Kenneth R., Howard J. Diamond, James P. Kossin, and Michael C. Kruk. (2018). 'International best track archive for climate stewardship (IBTrACS) project, version 4', NOAA National Centers for Environmental Information, http://doi.org/10.25921/82ty-9e16 (last accessed: October 2019).
Knapp, Kenneth R., Michael C. Kruk, David H. Levinson, Howard J. Diamond, and Charles J. Neumann. (2010). 'The international best track archive for climate stewardship (IBTrACS) unifying tropical cyclone data', *Bulletin of the American Meteorological Society*, vol. 91(3), pp. 363–76.
Kousky, Carolyn. (2014). 'Informing climate adaptation: A review of the economic costs of natural disasters', *Energy Economics*, vol. 46, pp. 576–92.
Kousky, Carolyn. (2016). 'Impacts of natural disasters on children', *The Future of Children*, vol. 26(1), pp. 73–92.
Krebs, Tom, Moritz Kuhn, and Mark L.J. Wright. (2015). 'Human capital risk, contract enforcement, and the macroeconomy', *American Economic Review*, vol. 105(11), pp. 3223–72.
Krebs, Tom. (2003a). 'Growth and welfare effects of business cycles in economies with idiosyncratic human capital risk', *Review of Economic Dynamics*, vol. 6(4), pp. 846–68.
Krebs, Tom. (2003b). 'Human capital risk and economic growth', *The Quarterly Journal of Economics*, vol. 118(2), pp. 709–44.
Krebs, Tom. (2006). 'Recursive equilibrium in endogenous growth models with incomplete markets', *Economic Theory*, vol. 29(3), pp. 505–23.
Lemoine, Derek. (2021). 'The climate risk premium: How uncertainty affects the social cost of carbon', *Journal of the Association of Environmental and Resource Economists*, vol. 8(1), pp. 27–57.
Loayza, Norman V., Eduardo Olaberria, Jamele Rigolini, and Luc Christiaensen. (2012). 'Natural disasters and growth: Going beyond the averages', *World Development*, vol. 40(7), pp. 1317–36.
Lucas, Robert E., Jr. (1987). *Models of Business Cycles*, Hoboken, NJ, USA: John Wiley and Sons.
Lucas, Robert E., Jr. (1988). 'On the mechanics of economic development', *Journal of Monetary Economics*, vol. 22(1), pp. 3–42.
Manabe, Syukaro, Ronald J. Stouffer, Michael J. Spelman, and Kirk Bryan. (1991). 'Transient responses of a coupled ocean–atmosphere model to gradual changes of atmospheric CO2. Part I. Annual mean response', *Journal of Climate*, vol. 4(8), pp. 785–818.
Mankiw, N. Gregory, David Romer, and David N. Weil. (1992). 'A contribution to the empirics of economic growth', *The Quarterly Journal of Economics*, vol. 107(2), pp. 407–37.
Manuelli, Rodolfo E. and Ananth Seshadri. (2014). 'Human capital and the wealth of nations', *American Economic Review*, vol. 104(9), pp. 2736–62.
McDermott, Thomas K.J., Frank Barry, and Richard S.J. Tol. (2014). 'Disasters and development: natural disasters, credit constraints, and economic growth', *Oxford Economic Papers*, vol. 66(3), pp. 750–73.

© The Author(s) 2025.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

Mendelsohn, Robert, Kerry Emanuel, Shun Chonabayashi, and Laura Bakkensen. (2012). 'The impact of climate change on global tropical cyclone damage', *Nature Climate Change*, vol. 2(3), pp. 205–9.
Moore, Frances C. and Delavane B. Diaz. (2015). 'Temperature impacts on economic growth warrant stringent mitigation policy', *Nature Climate Change*, vol. 5(2), pp. 127–31.
Müller-Fürstenberger, Georg and Ingmar Schumacher. (2015). 'Insurance and climate-driven extreme events', *Journal of Economic Dynamics and Control*, vol. 54, pp. 59–73.
Narita, Daiju, Richard S.J. Tol, and David Anthoff. (2009). 'Damage costs of climate change through intensification of tropical cyclone activities: An application of FUND', *Climate Research*, vol. 39(2), pp. 87–97.
Neumayer, Eric, Thomas Plümper, and Fabian Barthel. (2014). 'The political economy of natural disaster damage', *Global Environmental Change*, vol. 24, pp. 8–19.
Newell, Richard G., Brian C. Prest, and Steven E. Sexton. (2021). 'The GDP-temperature relationship: Implications for climate change damages', *Journal of Environmental Economics and Management*, vol. 108, 102445.
Nordhaus, William D. (2010a). 'Economic aspects of global warming in a post-Copenhagen environment', *Proceedings of the National Academy of Sciences*, vol. 107(26), pp. 11721–26.
Nordhaus, William D. (2010b). 'The economics of hurricanes and implications of global warming', *Climate Change Economics*, vol. 01(01), pp. 1–20.
Nordhaus, William D. (2011). 'Estimates of the social cost of carbon: Background and results from the RICE-2011 model', Working Paper 17540, National Bureau of Economic Research.
Nordhaus, William D. (2017). 'Revisiting the social cost of carbon', *Proceedings of the National Academy of Sciences*, vol. 114(7), pp. 1518–23.
Noy, Ilan. (2009). 'The macroeconomic consequences of disasters', *Journal of Development Economics*, vol. 88(2), pp. 221–31.
Phan, Toan and Felipe Schwartzman. (2024). 'Climate defaults and financial adaptation', *European Economic Review*, vol. 170, 104866.
Pindyck, Robert S. and Neng Wang. (2013). 'The economic and policy consequences of catastrophes', *American Economic Journal: Economic Policy*, vol. 5(4), pp. 306–39.
Raddatz, Claudio E. (2009). 'The wrath of God: Macroeconomic costs of natural disasters', Working Paper 5039, World Bank.
Raddatz, Claudio. (2007). 'Are external shocks responsible for the instability of output in low-income countries?', *Journal of Development Economics*, vol. 84(1), pp. 155–87.
Ranson, Matthew, Carolyn Kousky, Matthias Ruth, Lesley Jantarasami, Allison Crimmins, and Lisa Tarquinio. (2014). 'Tropical and extratropical cyclone damages under climate change', *Climatic Change*, vol. 127(2), pp. 227–41.
Rubel, Franz and Markus Kottek. (2010). 'Observed and projected climate shifts 1901-2100 depicted by world maps of the Kóppen-Geiger climate classification', *Meteorologische Zeitschrift*, vol. 19(2), 135.
Schumacher, Ingmar and Eric Strobl. (2011). 'Economic development and losses due to natural disasters: The role of hazard exposure', *Ecological Economics*, vol. 72, pp. 97–105.
Skidmore, Mark and Hideki Toya. (2002). 'Do natural disasters promote long-run growth?', *Economic Inquiry*, vol. 40(4), pp. 664–87.
Strobl, Eric. (2011). 'The economic growth impact of hurricanes: Evidence from US coastal counties', *Review of Economics and Statistics*, vol. 93(2), pp. 575–89.
Tierney, Kathleen J. (1997). 'Business impacts of the Northridge earthquake', *Journal of Contingencies and Crisis Management*, vol. 5(2), pp. 87–97.
Van den Bremer, Ton S. and Frederick Van der Ploeg. (2021). 'The risk-adjusted carbon price', *American Economic Review*, vol. 111(9), pp. 2782–810.
WBWDI. (2019). 'World Development Indicators', The World Bank, https://datatopics.worldbank.org/world-development-indicators/ (last accessed: January 2025).
Wessel, Pål and Walter H.F. Smith. (1996). 'A global, self-consistent, hierarchical, high-resolution shoreline database', *Journal of Geophysical Research: Solid Earth*, vol. 101(B4), pp. 8741–43.

Downloaded from https://academic.oup.com/ej/advance-article/doi/10.1093/ej/ueaf050/8169904 by guest on 20 December 2025

© The Author(s) 2025.

The Economic Journal, 1–32 https://doi.org/10.1093/ej/ueaf050 © The Author(s) 2025. Published by Oxford University Press on behalf of Royal Economic Society. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com. Advance Access Publication Date: 20 June 2025